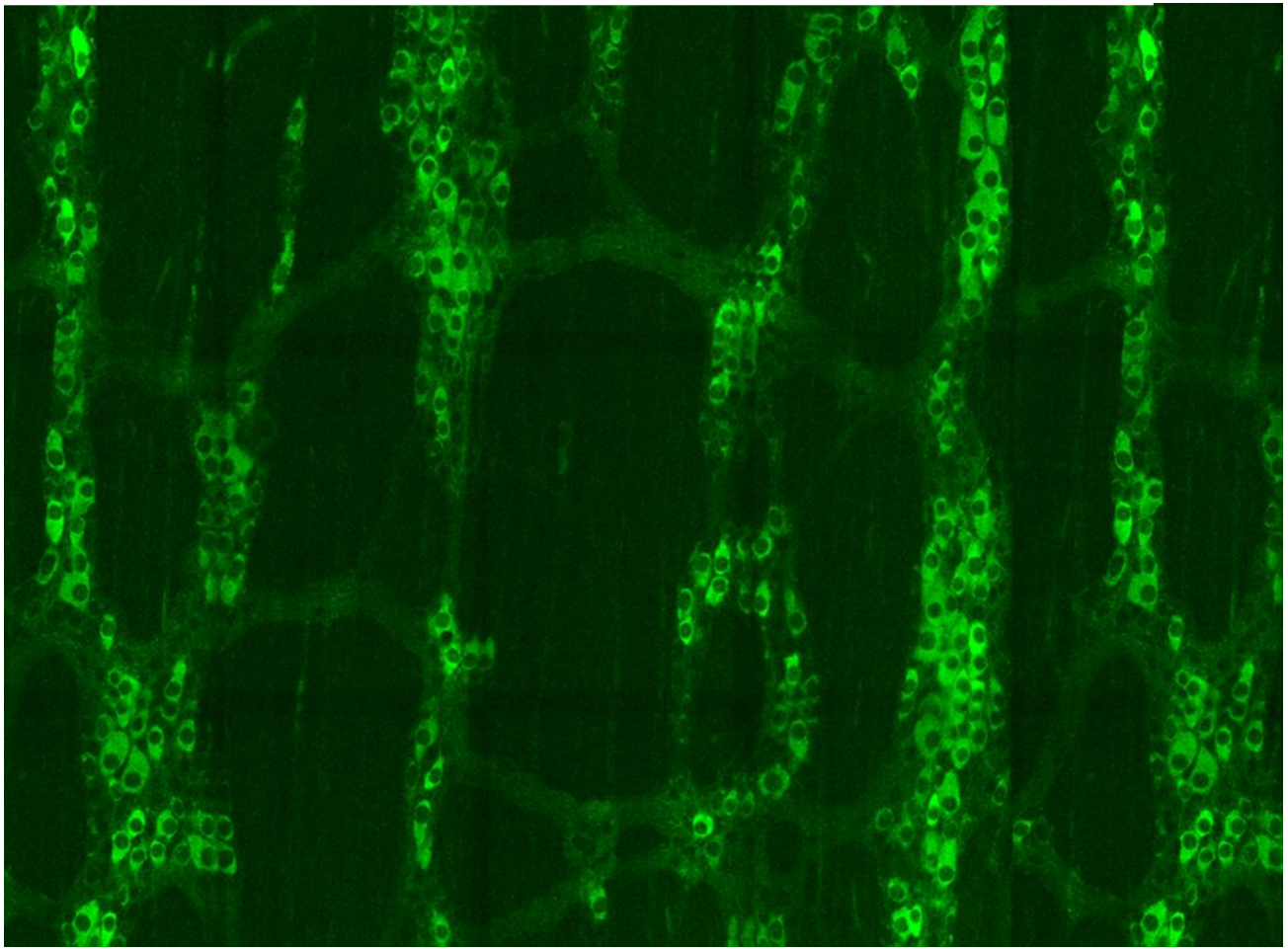
Does Gut Microbiome Dysbiosis Drive Gastrointestinal Neuropathy and Dysmotility in Type 2 Diabetes?

Onesmo B. Balemba, DVM, PhD

Biological Sciences &
WWAMI Medical Education Program

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Type 2 Diabetes (T2D)

- Chronic condition characterized by an inability to properly use or produce enough insulin, leading to hyperglycemia

A1C < 5.7
FBG < 100 mg/dL

Normal glucose tolerance
"Metabolically healthy"



A1C = 5.7-6.4
FBG = 100 - 125 mg/dL

Pre-diabetes (PD)



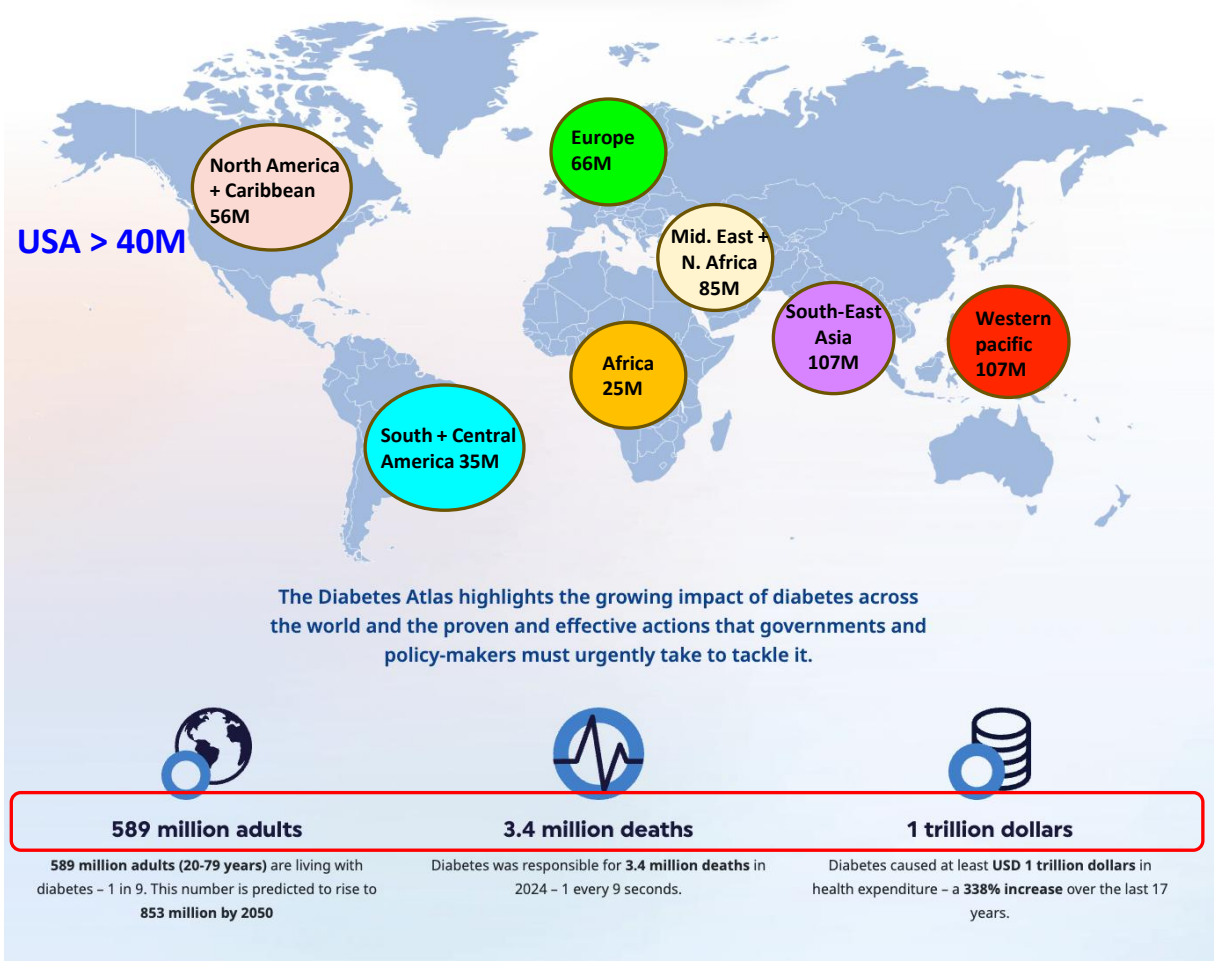
A1C ≥ 6.5%
FBG ≥ 126 mg/dL

Clinical Type 2 Diabetes (T2D)

Complications of T2D

https://diabetes.org/about-diabetes/diagnosis

T2D is a Growing Global Health Problem

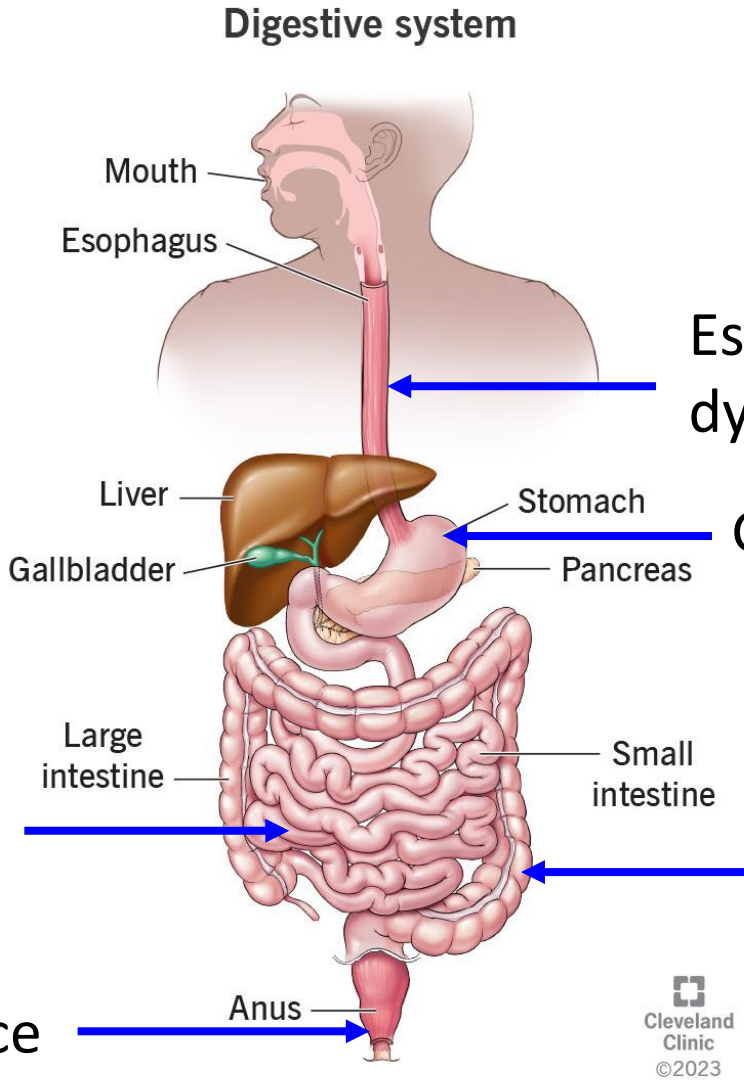


- > 541 million adults have prediabetes
- Yearly increase rate 13-134%
- 853 million by 2050

http://www.thelancet.com/diabetes-endocrinology https://idf.org/about-diabetes/diabetes-facts-figures/

Gastrointestinal (GI) Disorders in T2D Patients

- Up 76% of T2D patients
- Common among prediabetics and the obese



Esophageal dysmotility

Gastroparesis

Small intestinal bacterial overgrowth

Constipation/ Diarrhea/pain

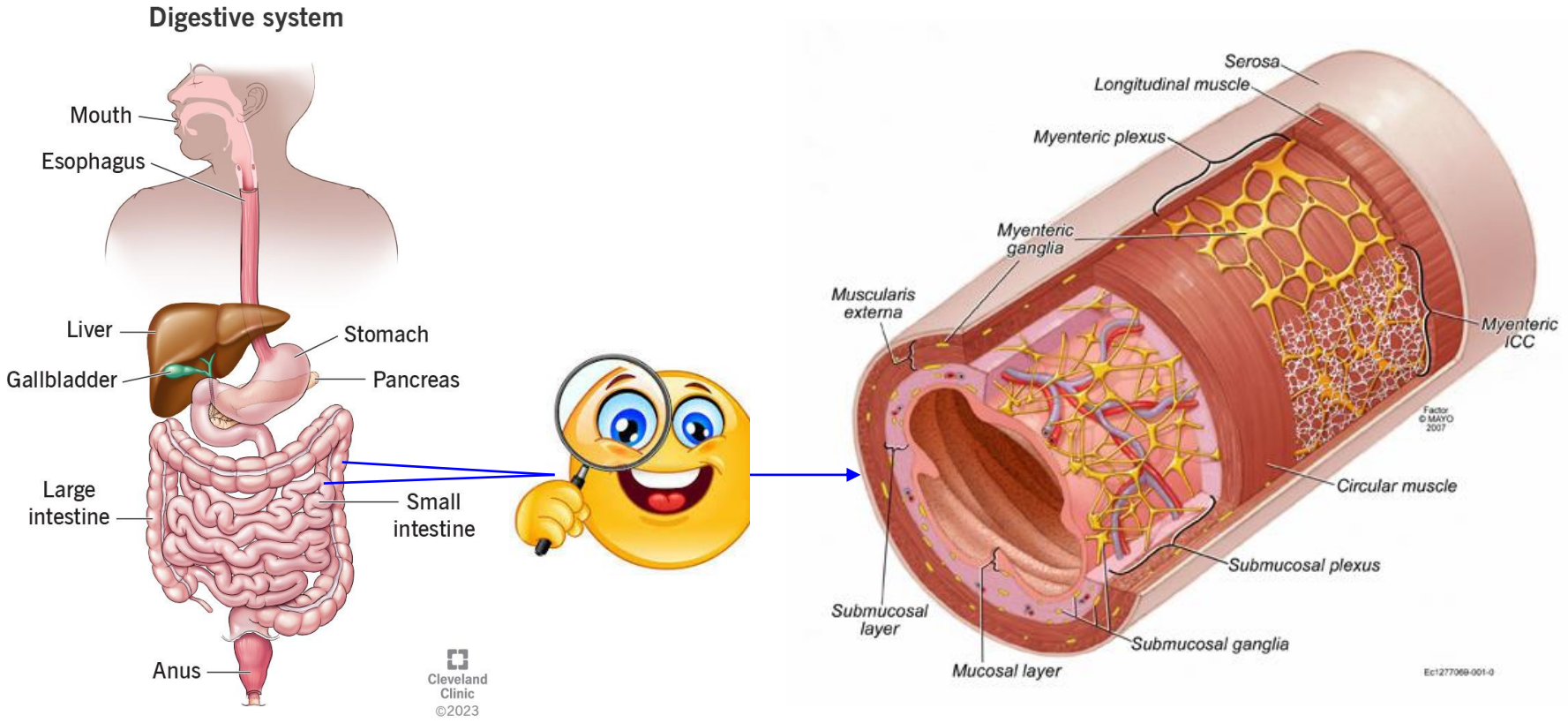
Fecal incontinence

https://doi.org/10.1016/j.ejps.2019.04.013

Causes of GI Disorders in Patients with T2D

- **Leaky gut and inflammation**
- **Damage to the enteric nervous system (ENS, enteric neuropathy)**
- **Damage to the muscular apparatus:**
  - Smooth muscle cells
  - Pacemaker interstitial cells of Cajal
  - Platelet-Derived Growth Factor positive cells

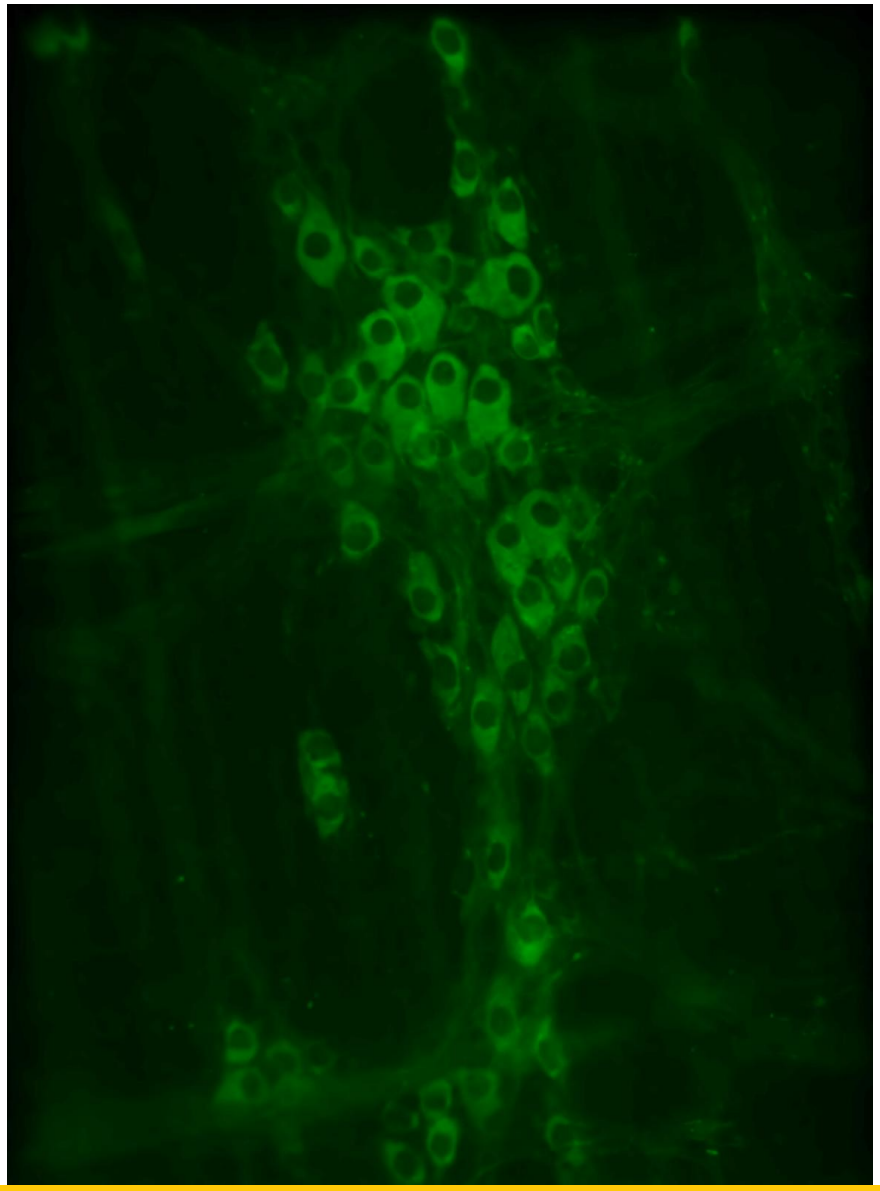
Enteric Nervous System (ENS, “Gut Brain”)



- > 600 million internal neurons and billions of enteric glial cells
- External sensory, parasympathetic, and sympathetic nerve fibers

PMID: 24997029 DOI: 10.1007/978-1-4939-0897-4_3

The Neuromuscular Apparatus



The ENS Regulates Gut Motility

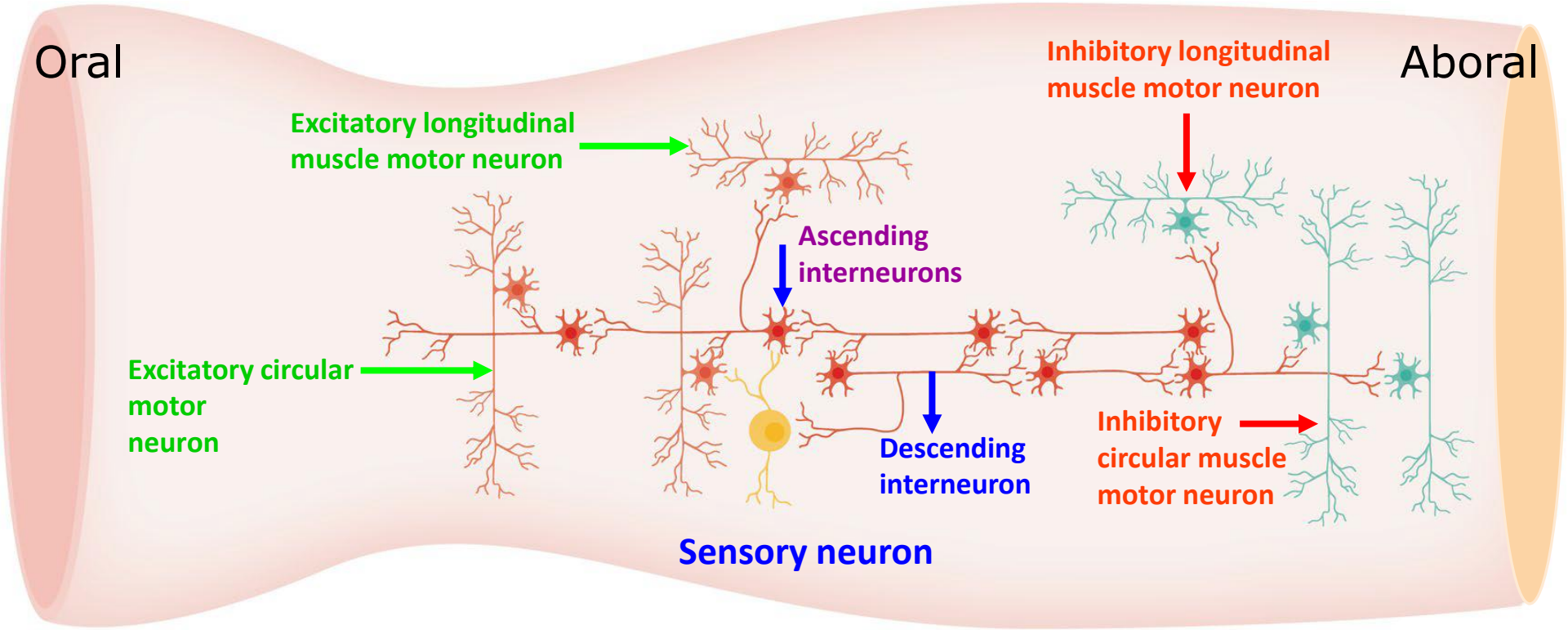
Oral  Aboral

Oral  Aboral

- Stimulus at any point in the GI triggers ENS reflexes that induce muscle oral contraction and aboral relaxation (Bayliss and Starling, 1899)

Journal of Physiology 1989 (411) pp 1-25

ENS Reflexes Regulate Gut Movement



- ENS neuronal reflexes involve sensory (detection), intermediate (integration), and motor neurons (actions) that drive gut motility and other functions via complex circuits

Spencer, N.J., Hu, H. Nat Rev Gastroenterol Hepatol 17, 338–351 (2020)

What causes ENS Neuropathy?

❖ **As a complication of T2D**

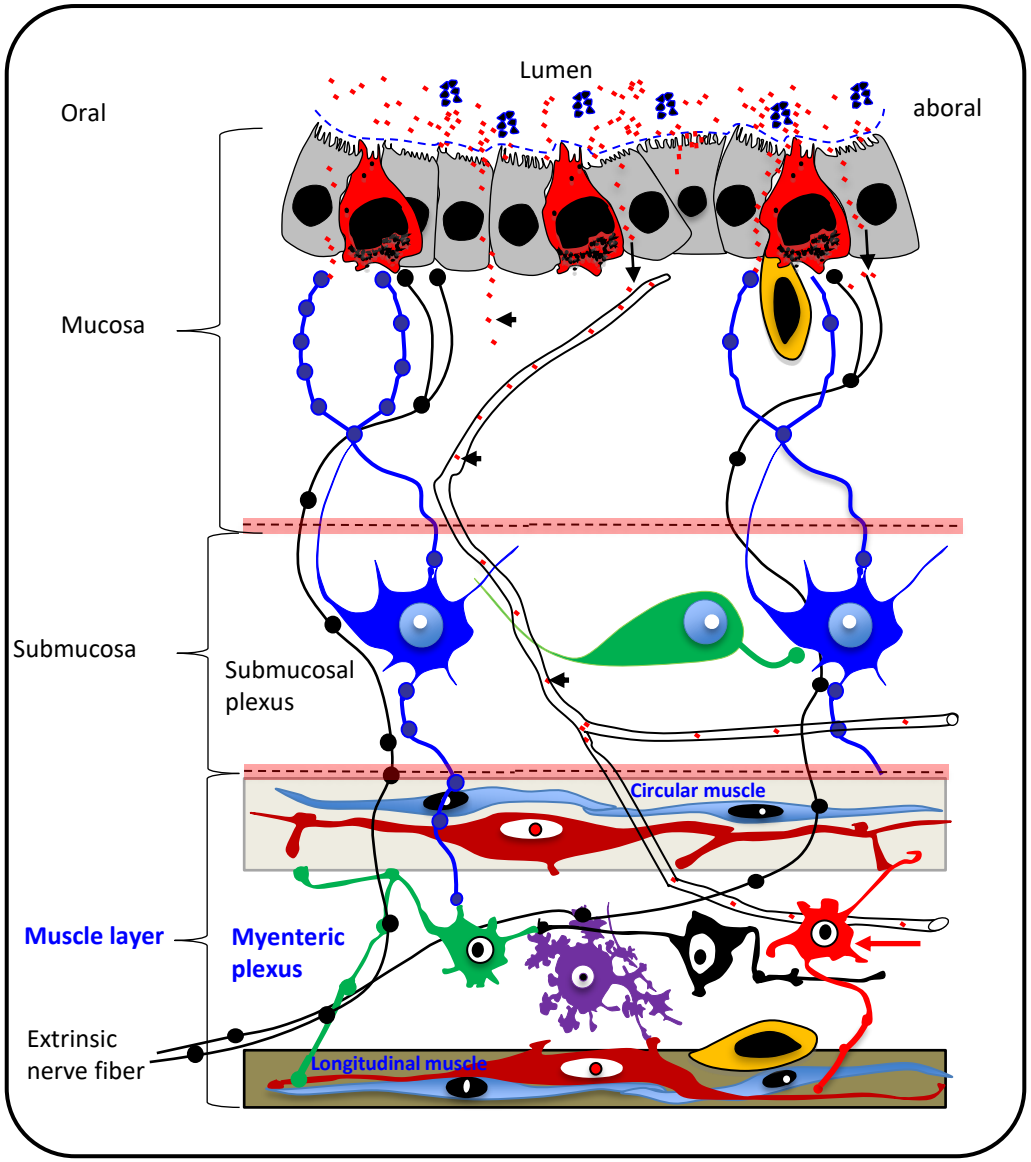
- Chronic hyperglycemia
- Chronic dyslipidemia
- Oxidative stress and metabolic injury
- Altered gut microbial community structure and function (dysbiosis of the microbiome)

❖ **Neuropathy occurs among prediabetics and the obese**

Our Central Hypothesis

- **Questions**:
  - Do ENS neuropathy and dysmotility precede T2D?
  - If so, what triggers them?

- **Hypothesis**: Toxic substances from gut contents trigger damage and dysfunction of the ENS and muscle tissue well before T2D



Research Objectives:

Determine whether molecules derived from diet-microbiota-host interactions (solubilized in fecal fluid) trigger diabetic ENS neuropathy and dysmotility

Determine whether an altered microbiome plays a role in the pathophysiology

High-Fat Diet (HFD)- induced ENS Neuropathy

8-week-old male & female C57BL6/J mice





6.5% kcal from fat - standard chow diet (SCD)

45-72% kcal from fat (high-fat diet, HFD)

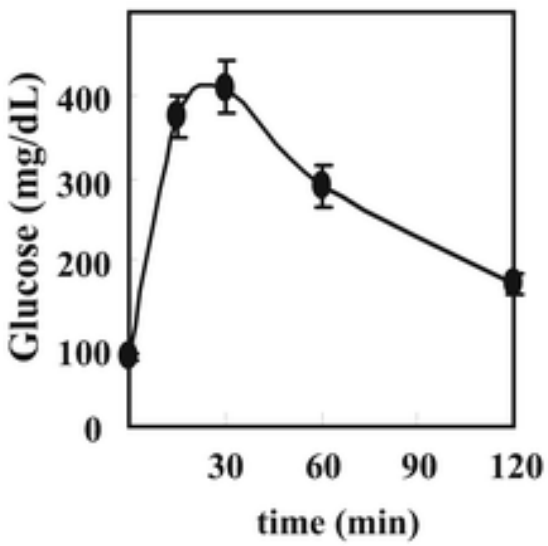
After 2- 20 weeks

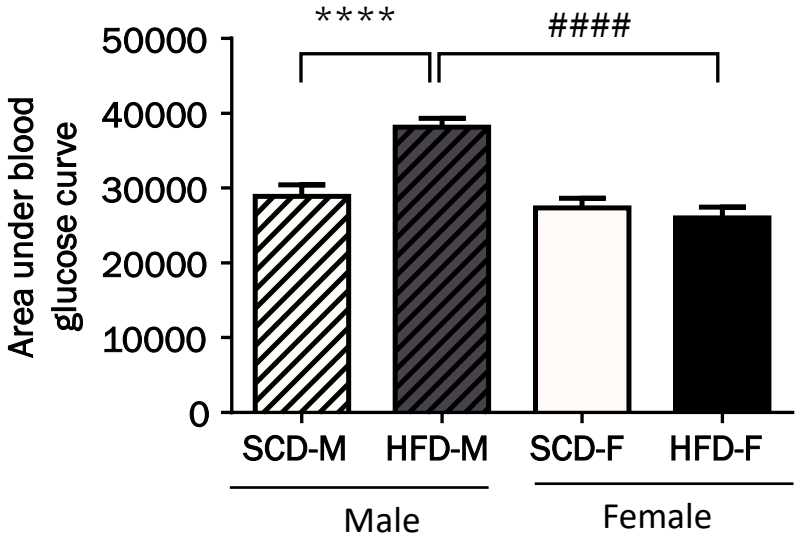


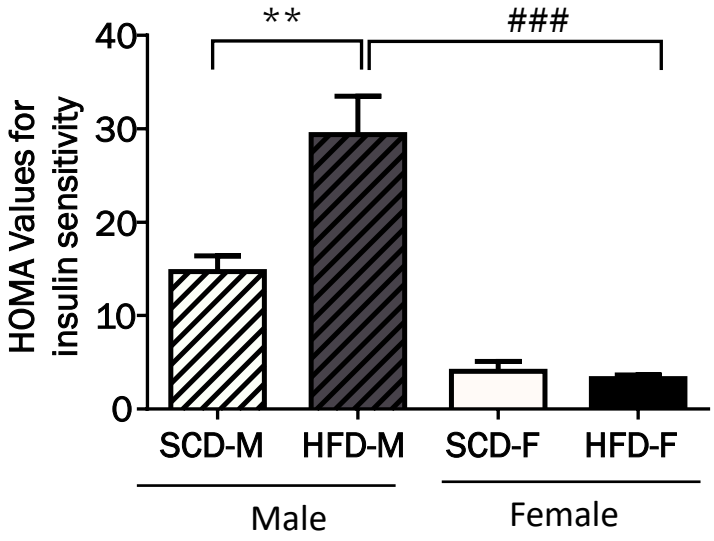


http://jaxmice.jax.org/strain/100006.html

Sex Matters: Sex-Related Metabolic Differences

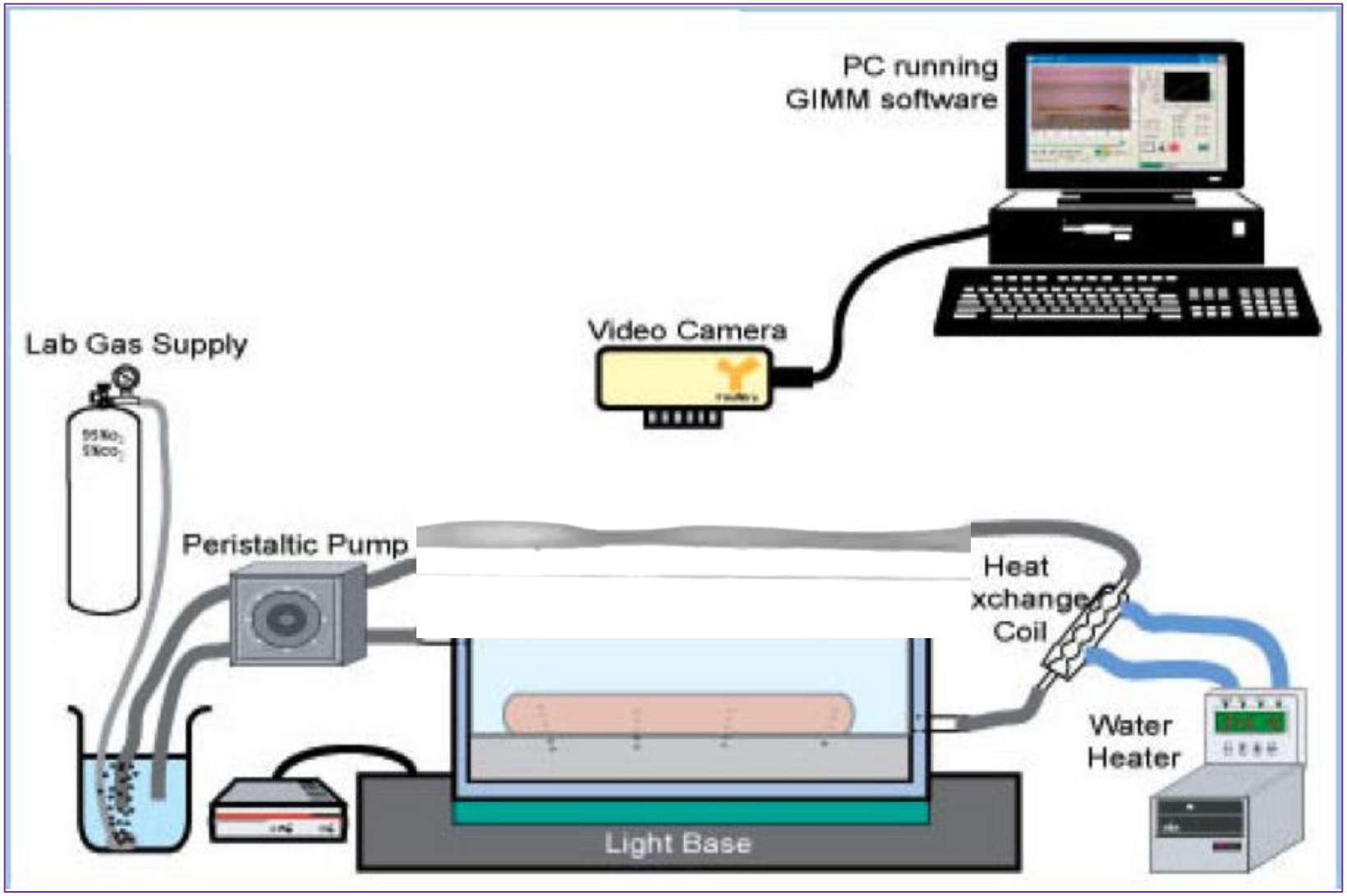






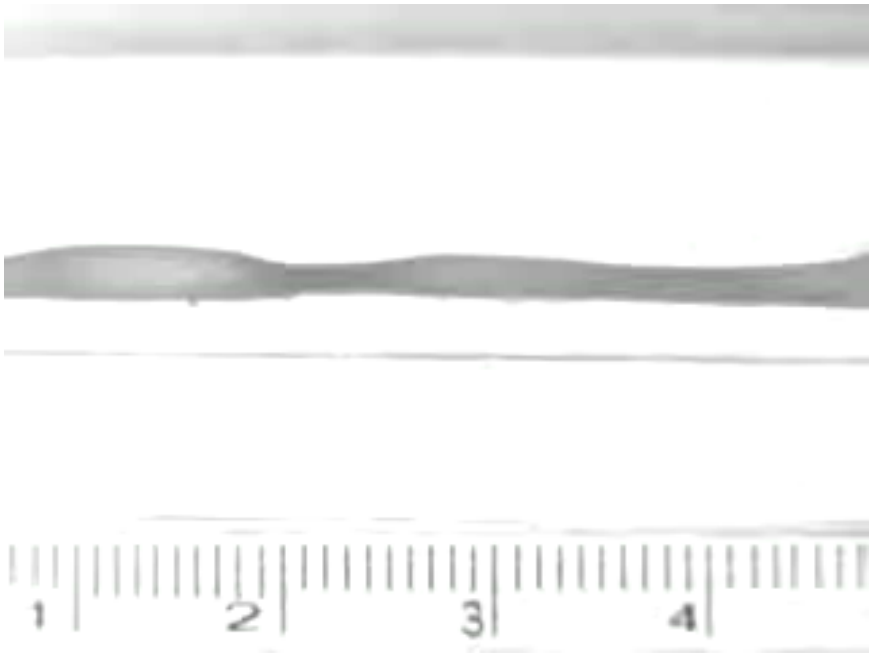
- An 8-week 72% kcal HFD regimen caused diabetic conditions in male but not female mice

Does HFD Disrupt Bowel Movement?

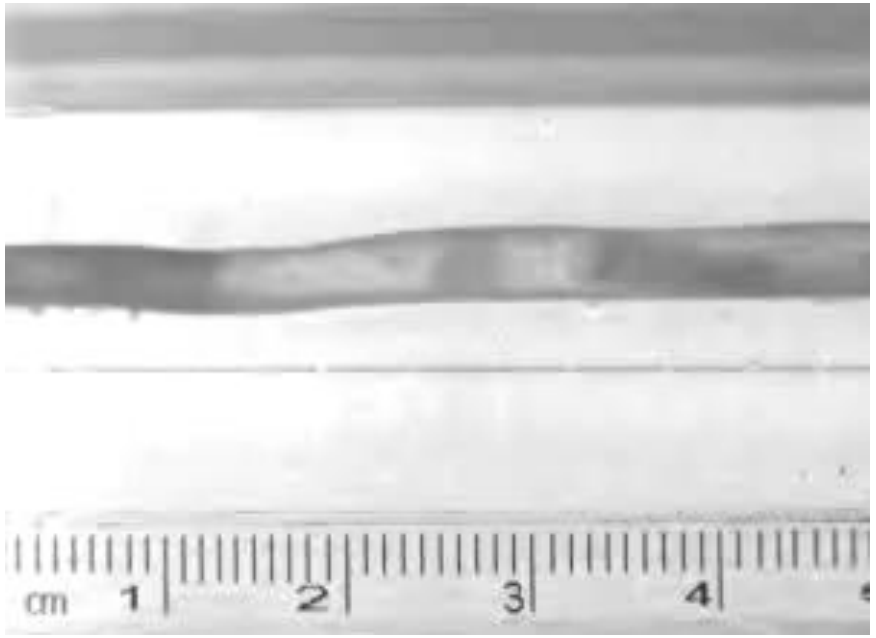


In vitro gastrointestinal motility monitor system

HFD Impairs Small and Larger Bowel Motility



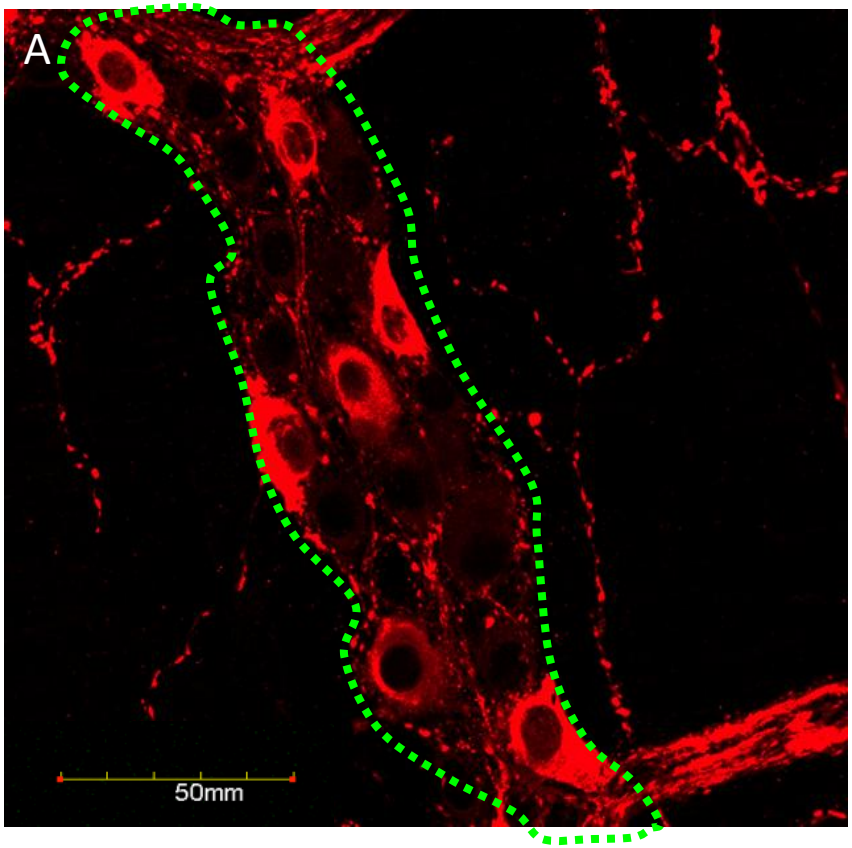
Regular chow female mouse

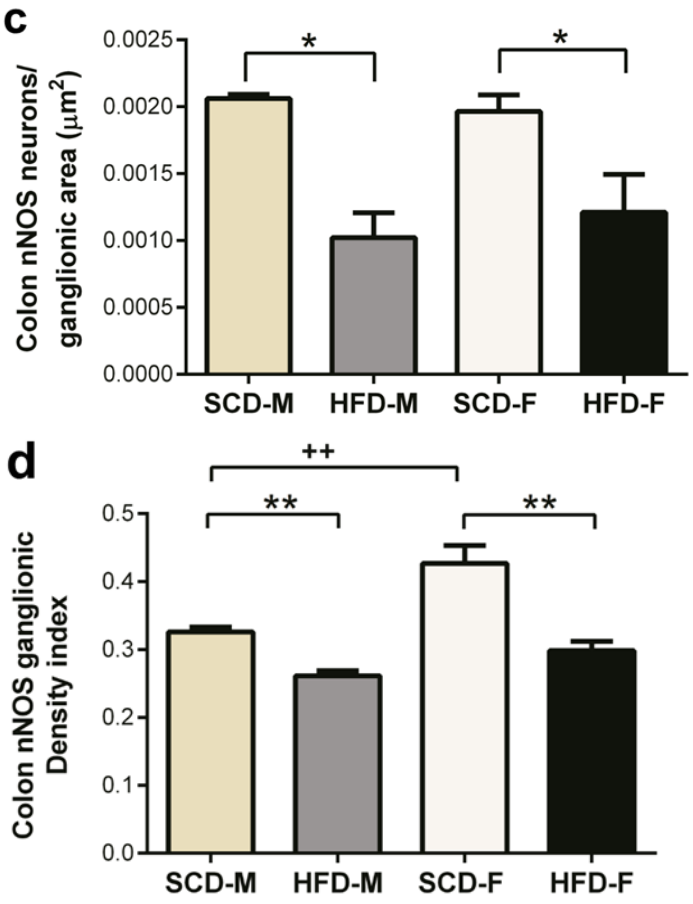


8-week 72% kcal HFD female mouse

HFD causes Damage to the ENS and Dysmotility Before Glucose Intolerance

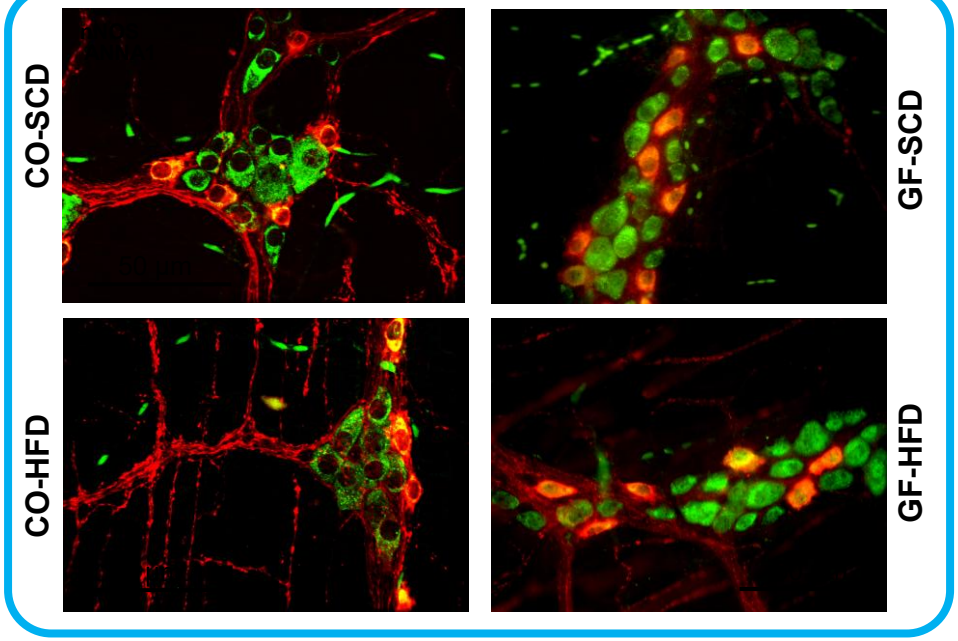
Nitrergic (nNOS) myenteric neurons, colon





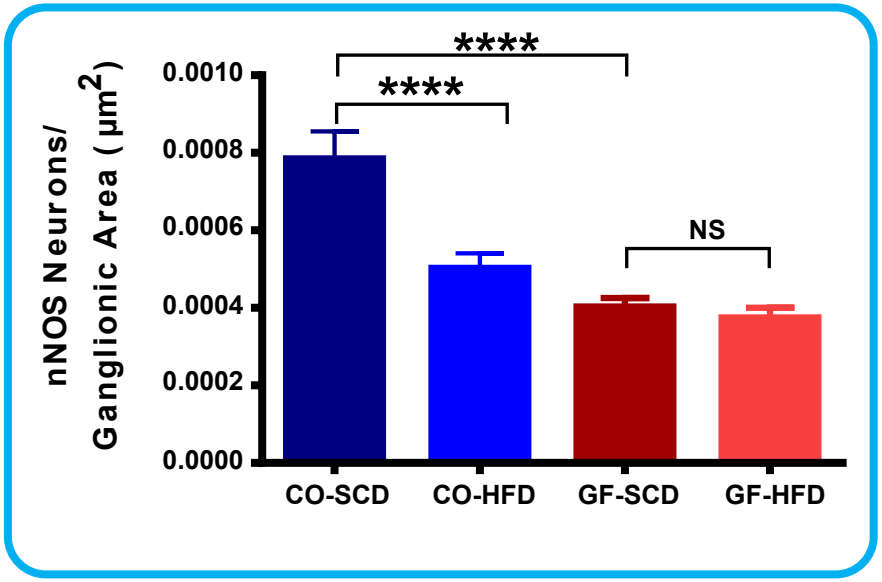
nNOS = neuronal nitric oxide synthase

Do the Microbiota Contribute to HFD-Induced ENS Neuropathy and Dysmotility?



Red = nNOS for nitrergic (inhibitory motor neurons)

Green= ANNA1 antibody for all neurons

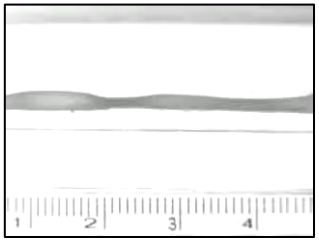


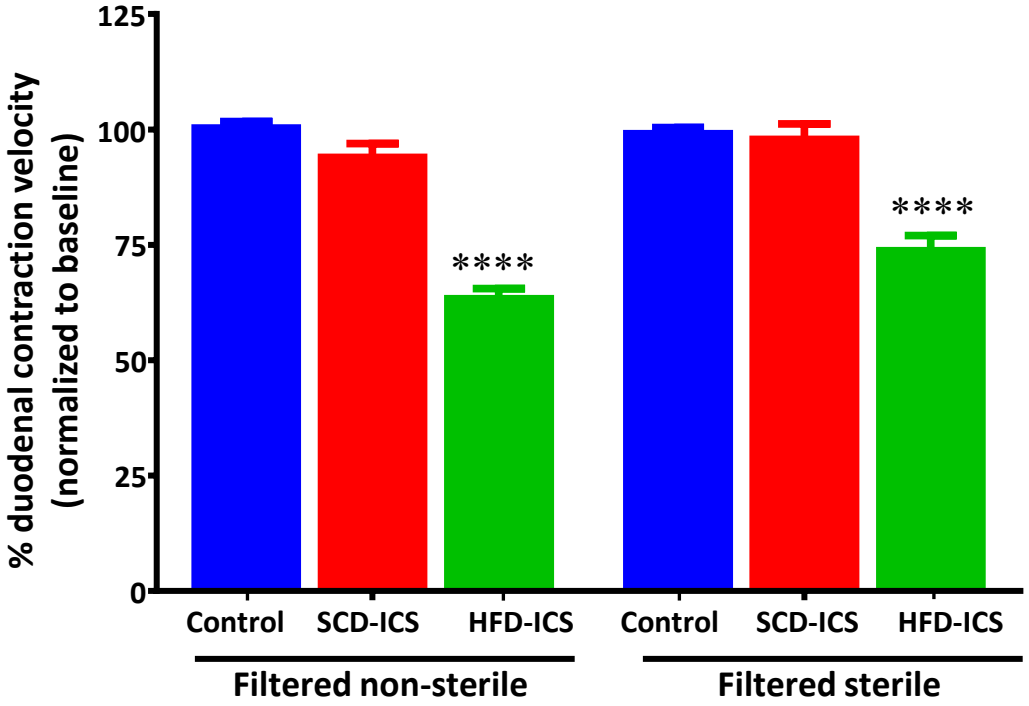
CO = Conventional mice

GF = Germ-free mice

- Germ-free mice are protected against HFD-induced injury to ENS neurons

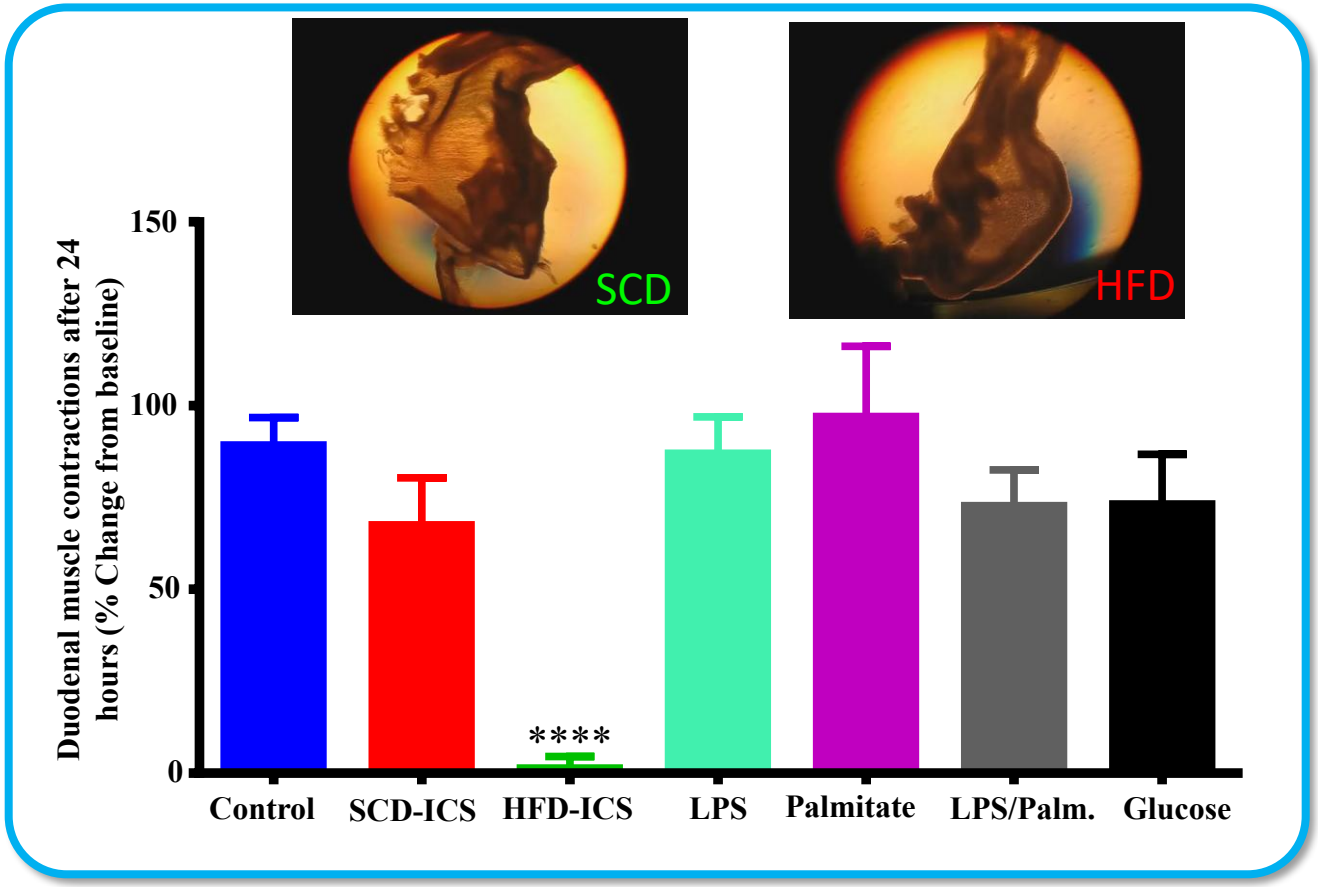
Do Molecules in the Fecal Fluids of HFD Mice Trigger Dysmotility *Ex Vivo*?





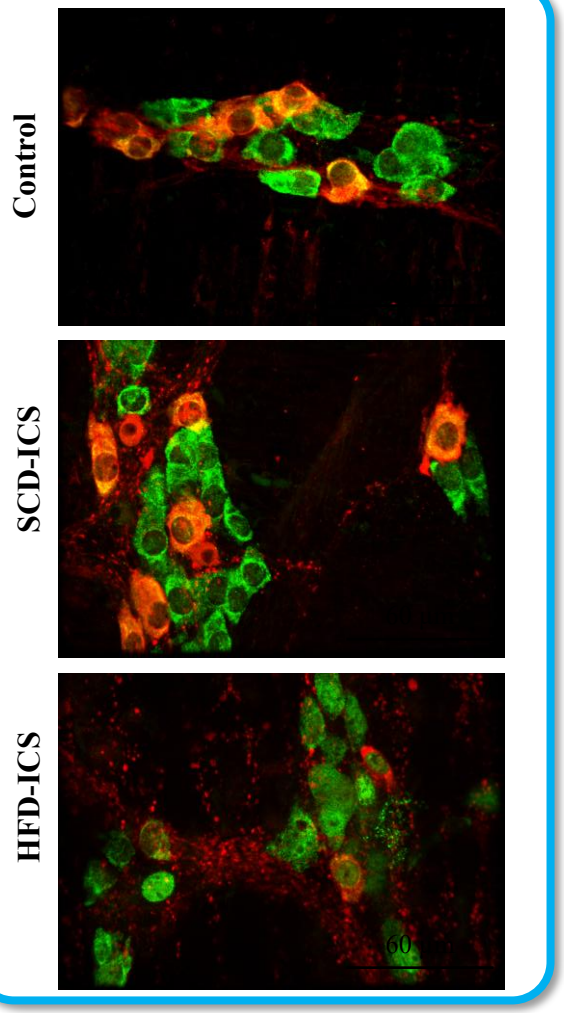
- Fecal filtrates (supernatants, ICS) from 8-week HFD male (diabetic) and female (non-diabetic) mice decreased propulsive movements in isolated duodenums and colons

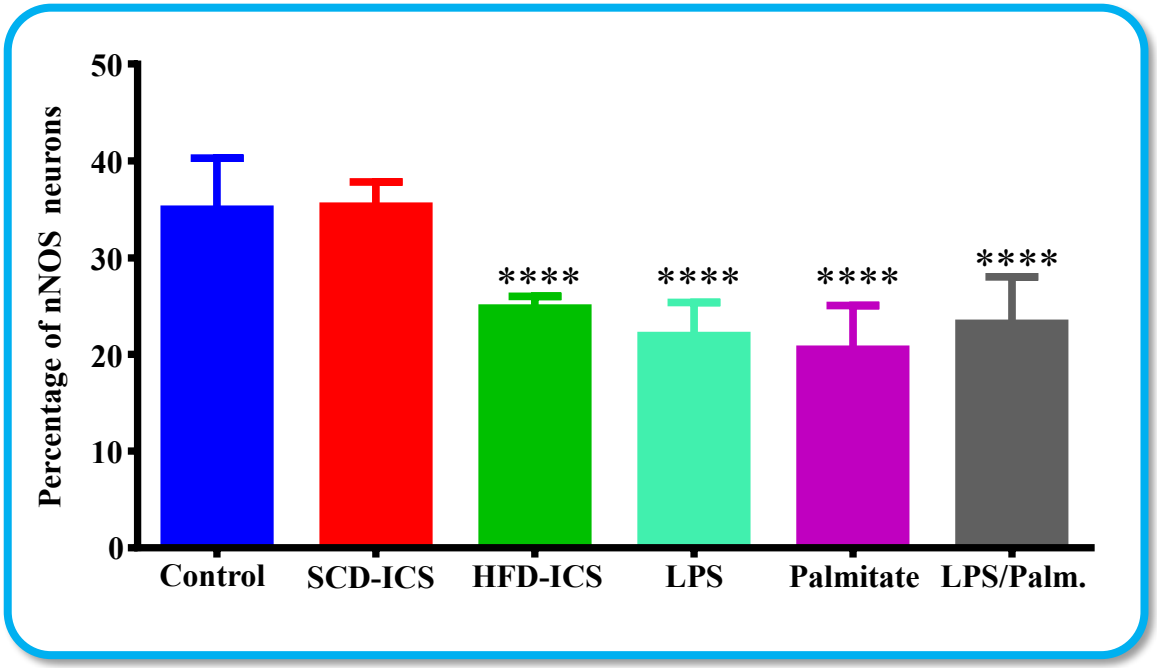
Do Molecules in the Fecal Fluids of HFD Mice Impair Intestinal Muscle Contractions?



- Filter-sterilized fecal filtrates (ICS) from HFD male and female mice decreased muscle contractions

Do Molecules in the Fecal Fluids of HFD Mice Damage Myenteric Neurons?

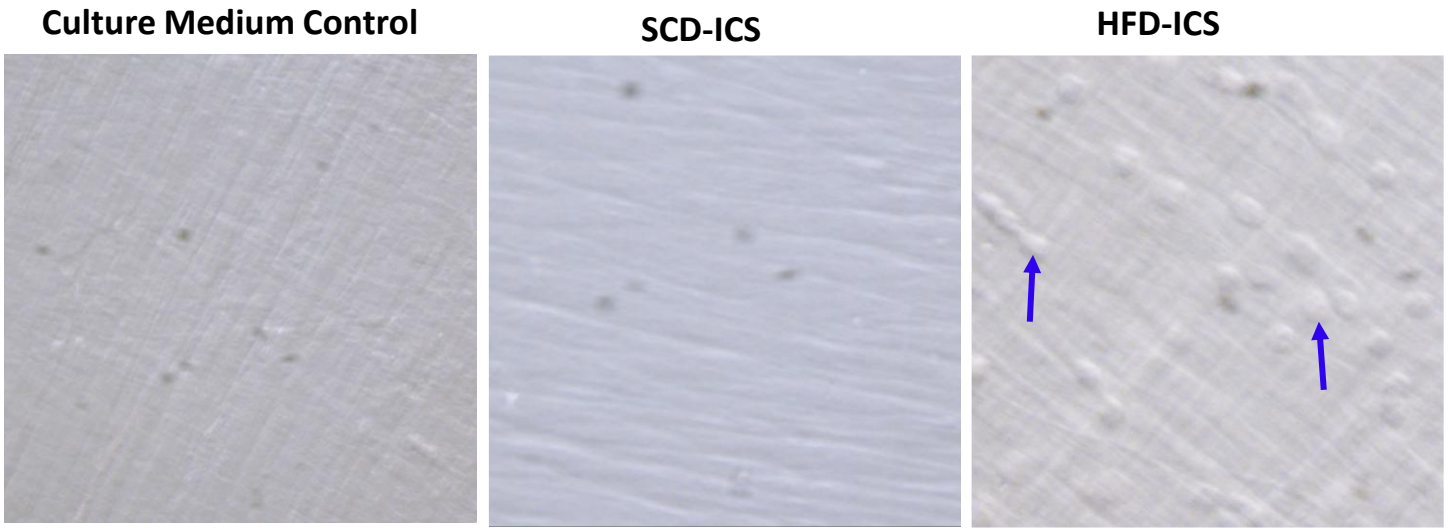


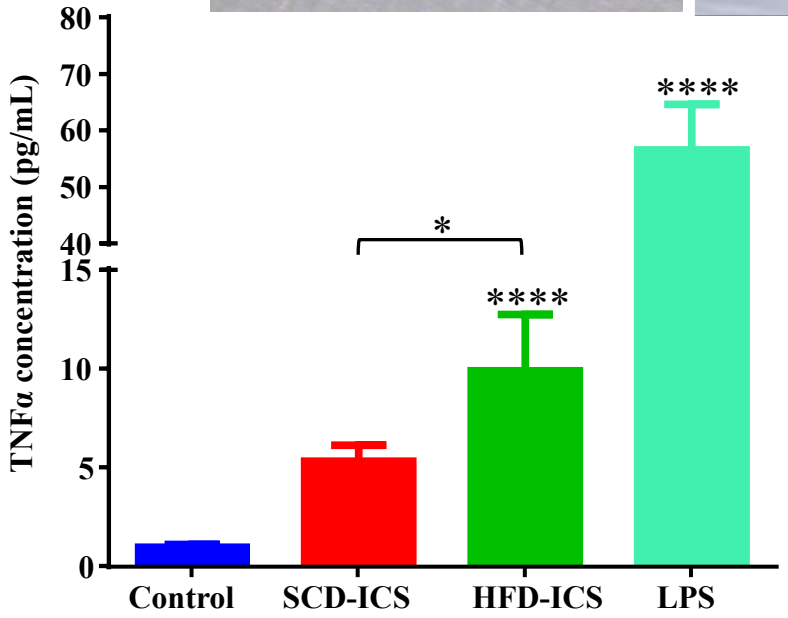


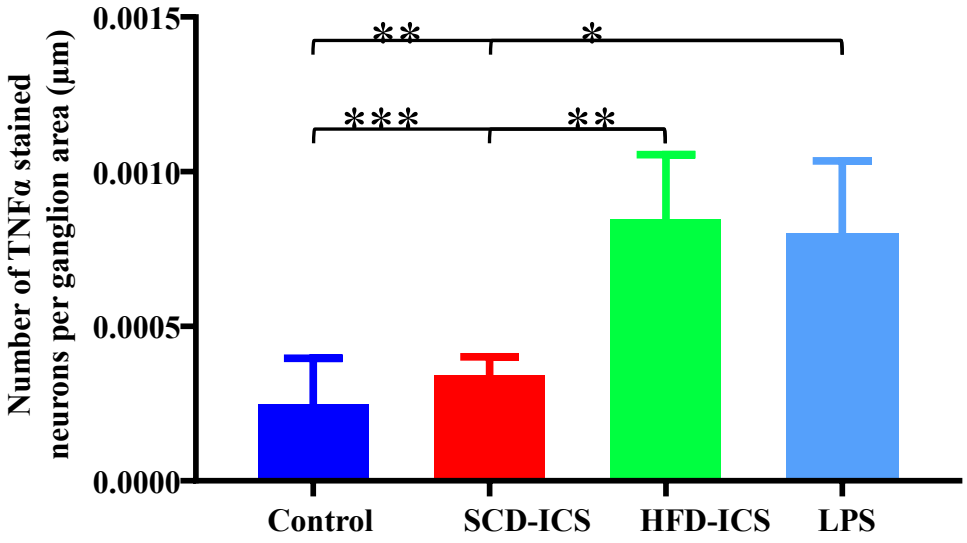
- Filter-sterilized fecal filtrates (ICS) from male and female HFD mice decreased the number of nitrergic (nNOS) myenteric neurons

Nyavor Y, et al., Neurogastroenterol Motil. 2021 Jan;33(1):e13990

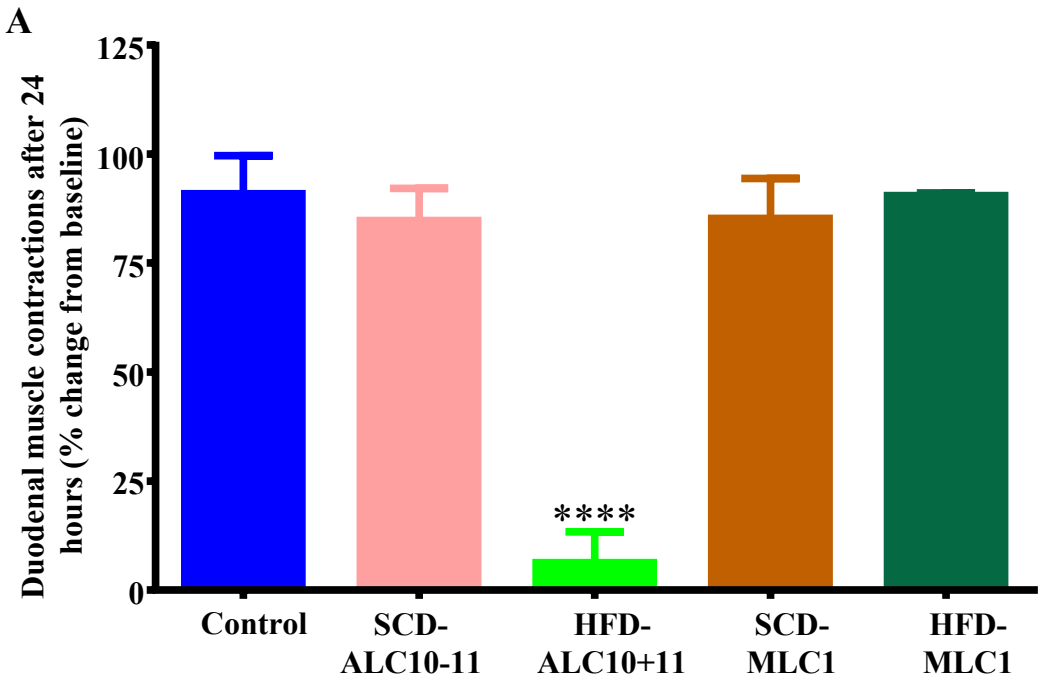
HFD Mice Fecal Supernatants Activate Inflammation

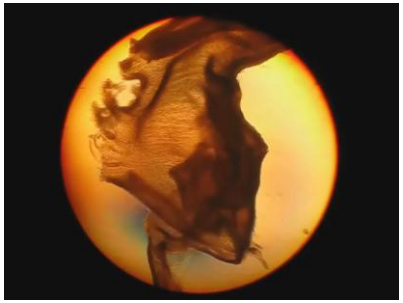


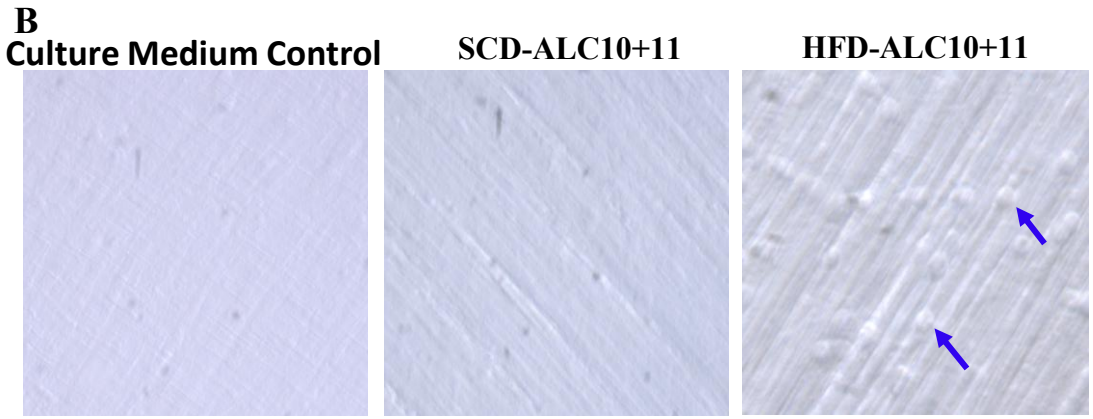




What Are Toxic Molecules?

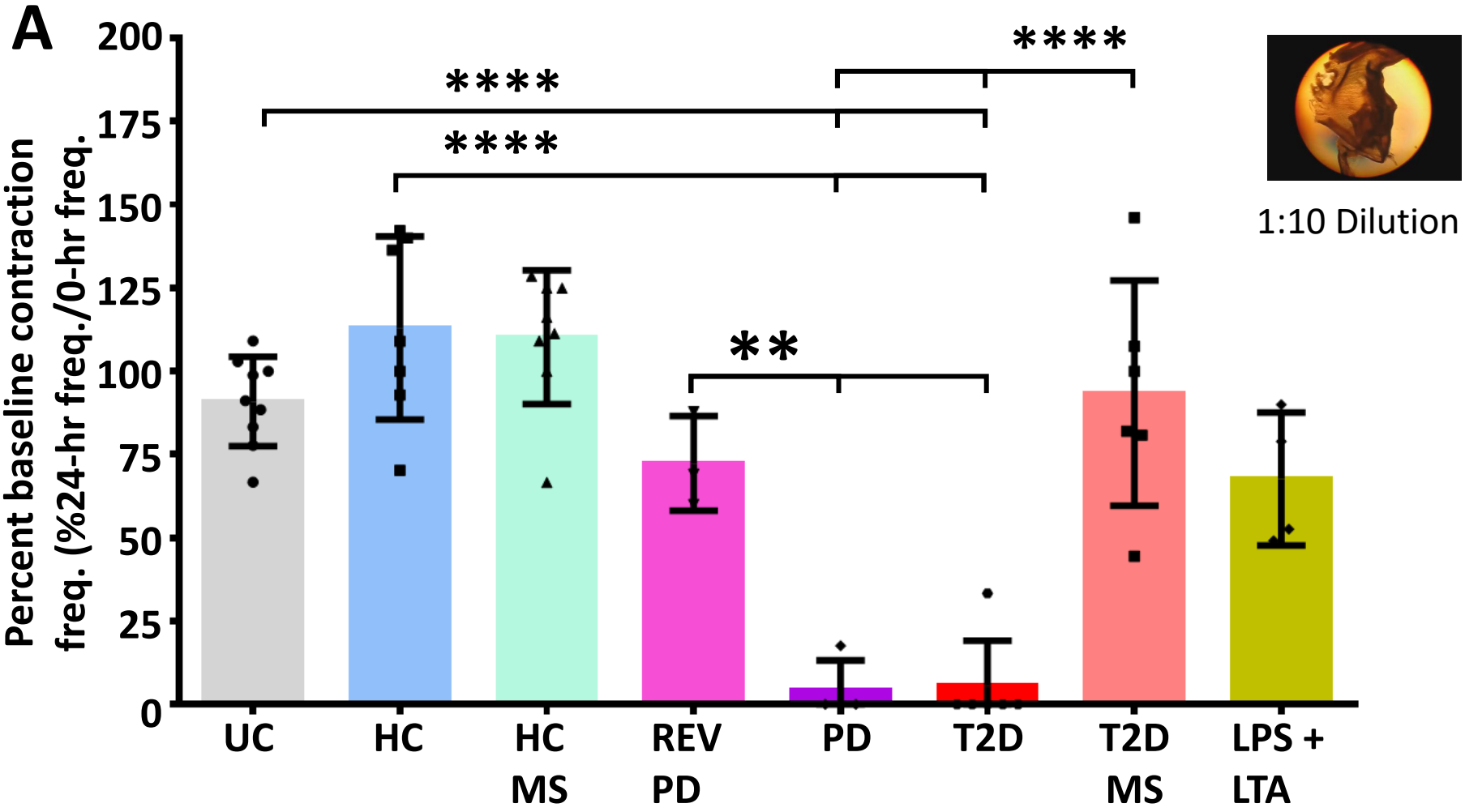






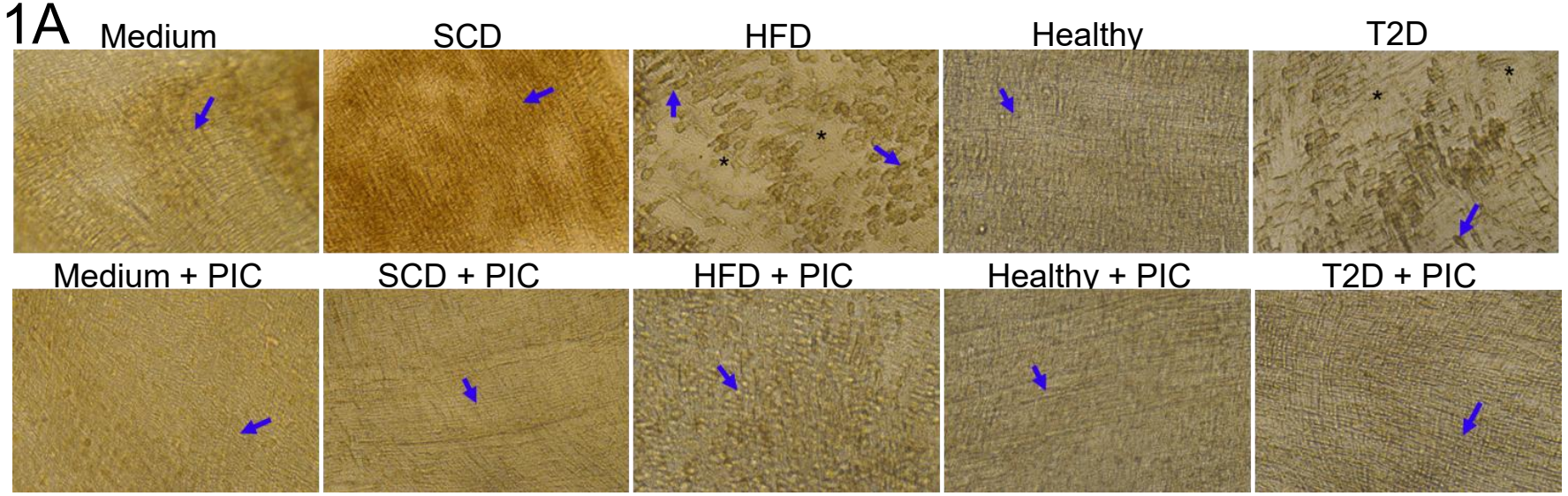
- Toxic molecules in stool supernatants are water-soluble and can be concentrated by HPLC

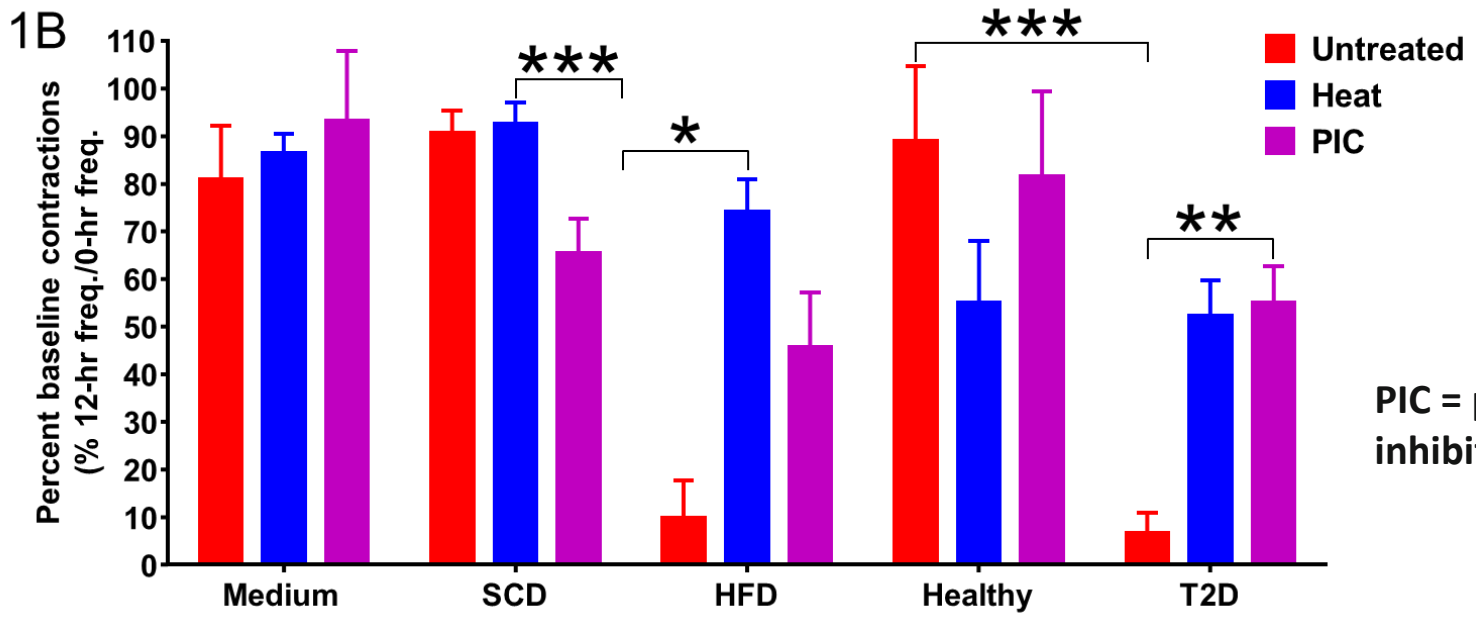
Do Mouse Findings Translate to Humans?



UC= Untreated, HC= Healthy, HC MS = Healthy "endotoxin" filtered out, REV PD= Reversed prediabetes, PD= prediabetes, T2D= type 2 diabetes, T2DMS= T2D "endotoxin" filtered out and LPS + LTA (endotoxin mixed)

Are the Toxic Molecules Proteases?

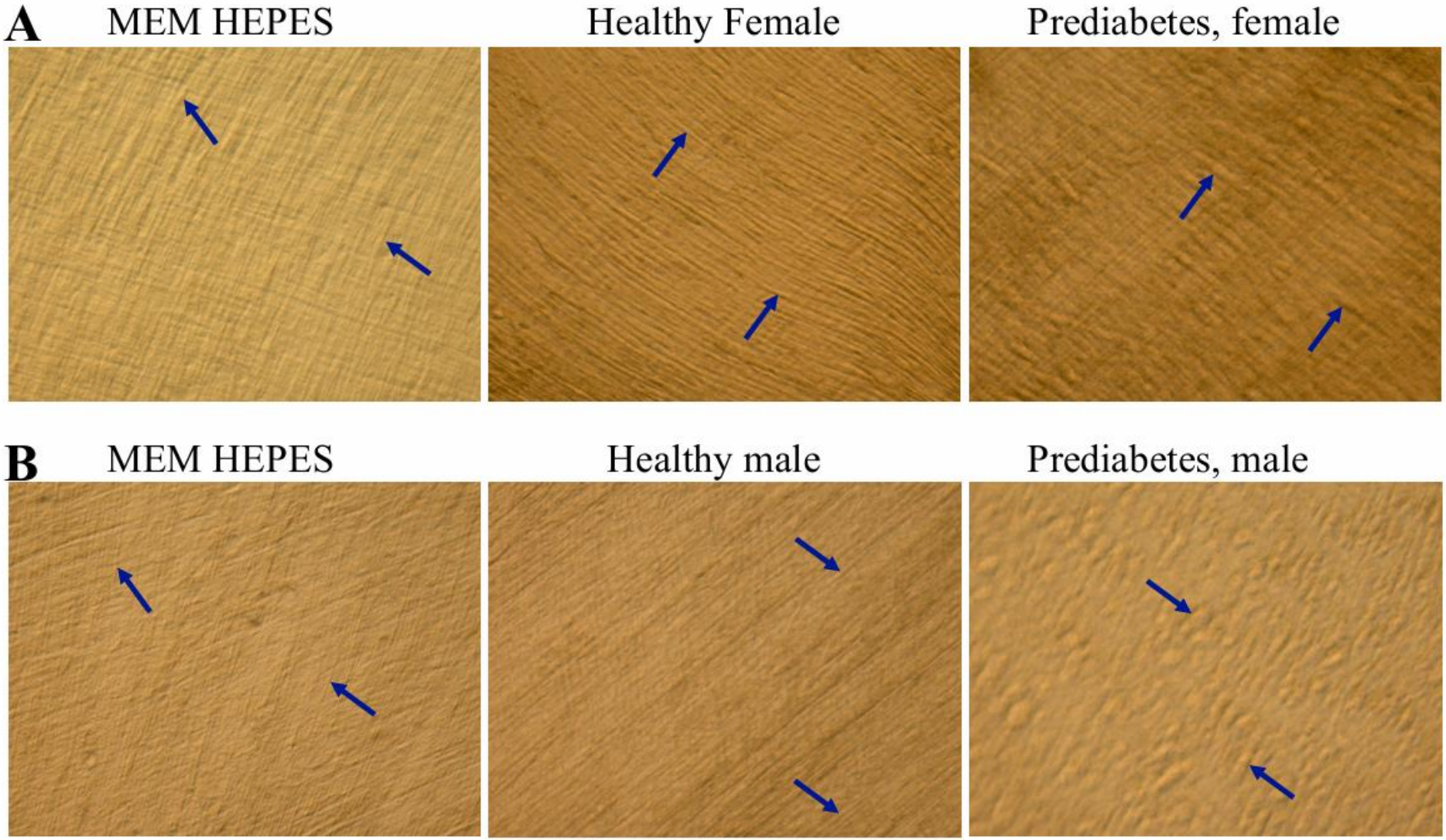




PIC = protease inhibitor cocktail

Voller, et al., Neurogastroenterol Motil. 2026 Jul;38(7):e70401. doi: 10.1111/nmo.70401

Are the Toxic Molecules Proteases?

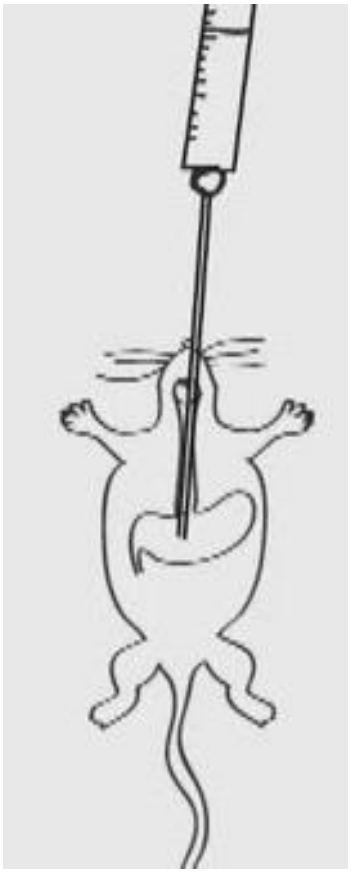


Do Fecal Fluid From Patients with T2D Impair ENS Neurons *in vivo?*

Fecal filtrates were transplanted into sex-matched healthy mice by gavage

1:10 dil.
X3 @ week
8 weeks



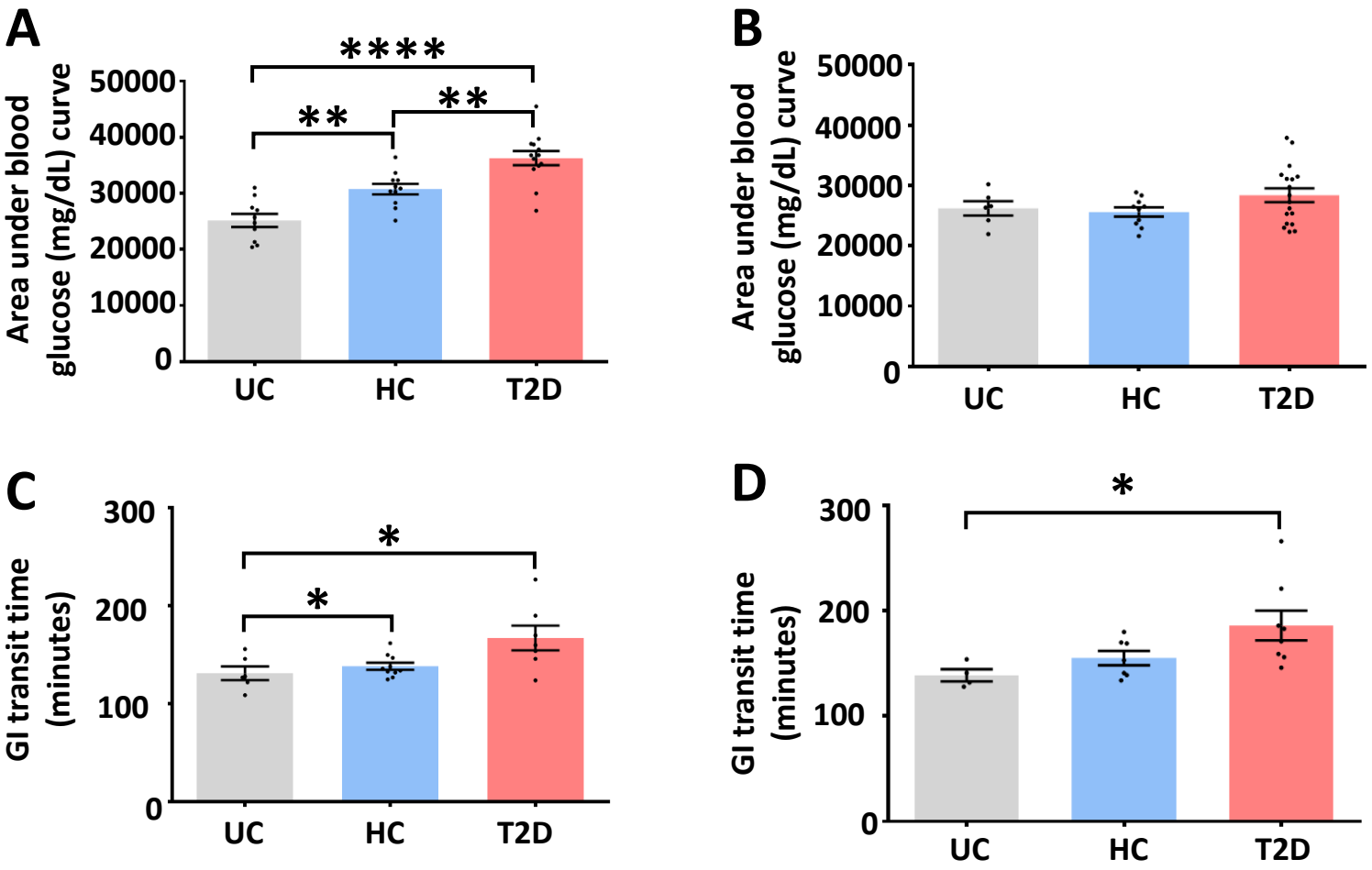


Control: Untreated or saline

Sex Matters: Sex-Related Metabolic Differences

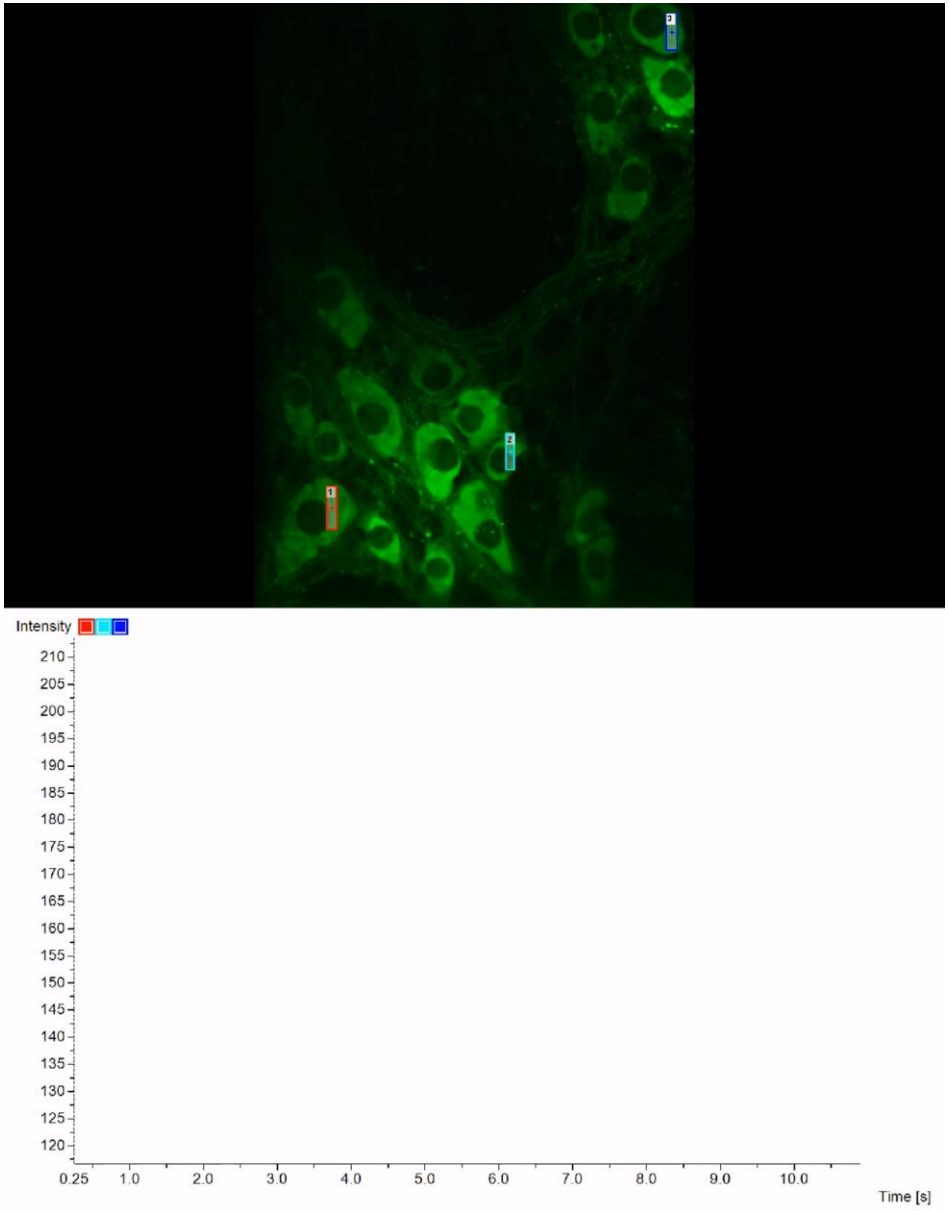
A & C: Male Recipients

B & D: Female recipients



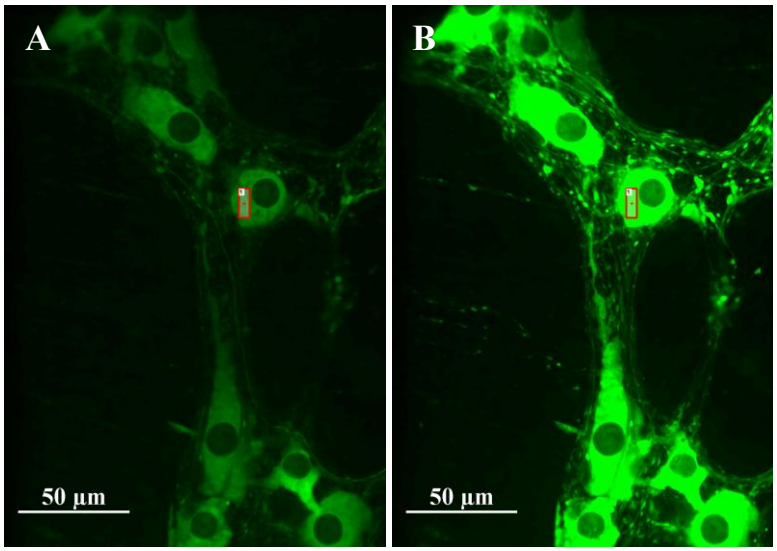
Voller, et al., Neurogastroenterol Motil. 2026 Jul;38(7):e70401. doi: 10.1111/nmo.70401

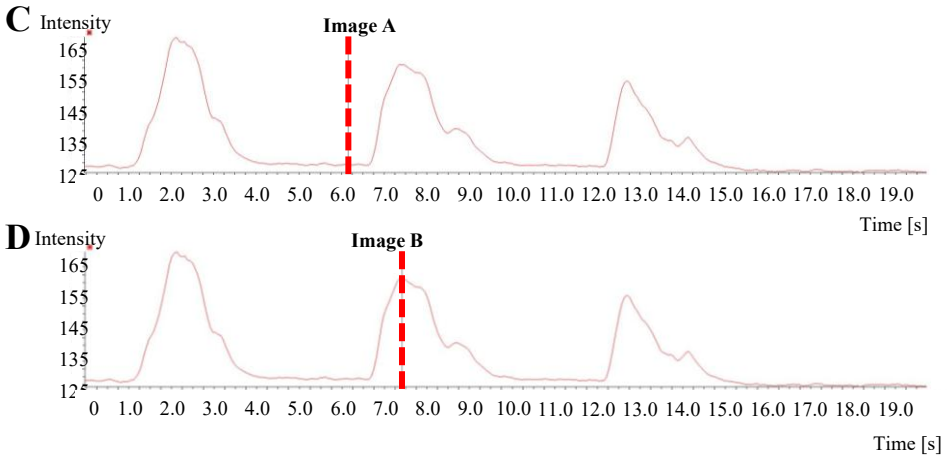
Effect on the Activity of Cholinergic Neurons



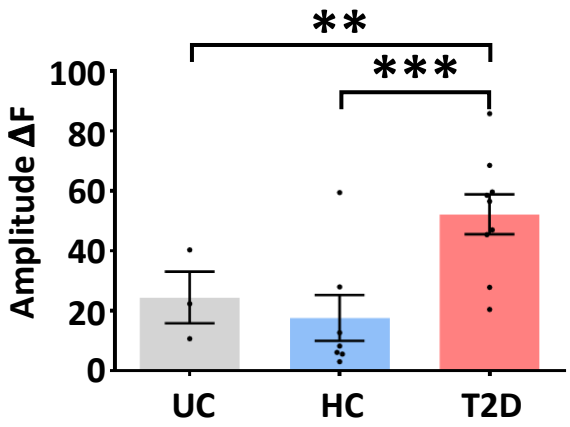
- Ca^{2+} binding protein GCaMP6f in cholinergic myenteric neurons

Sex Matters: Sex-Related Neuroactivity Differences

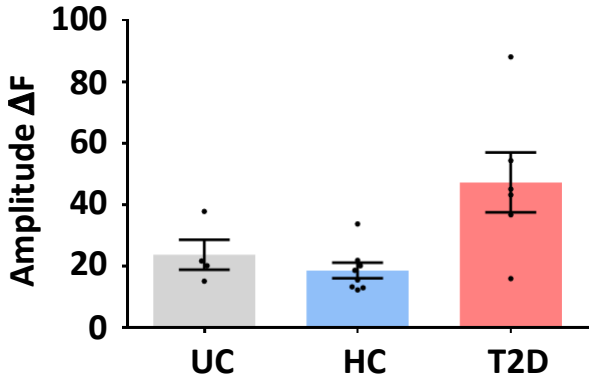




Male Recipients

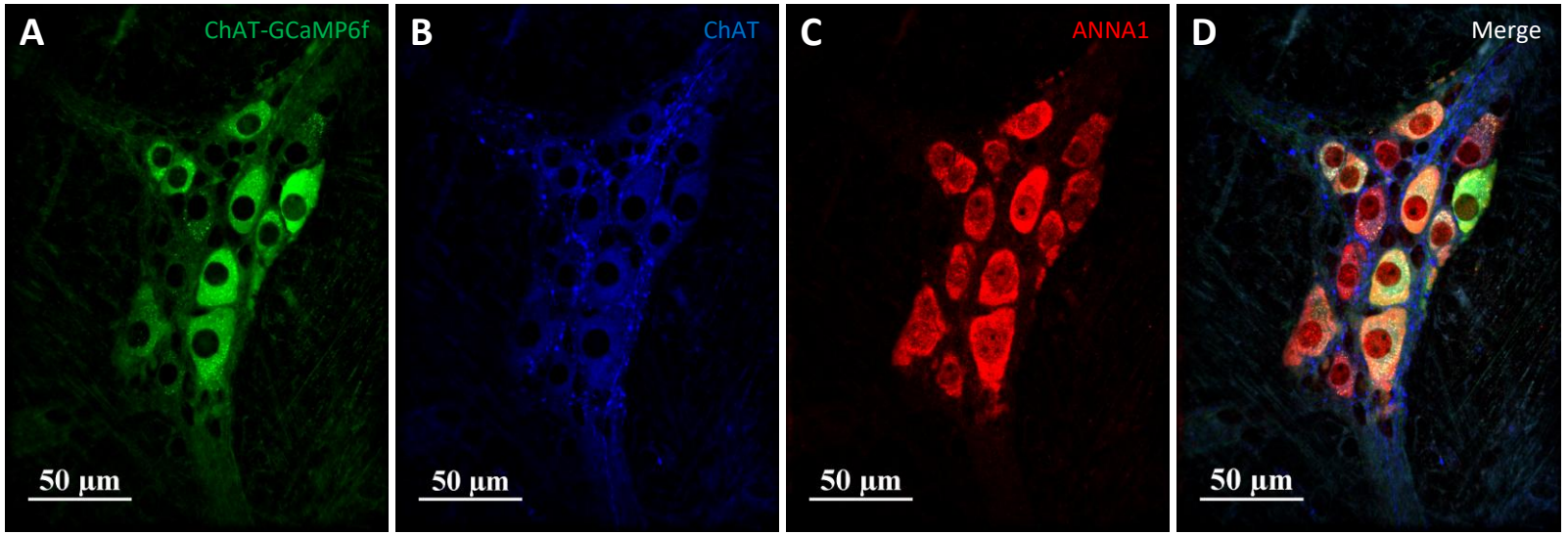


Female recipients

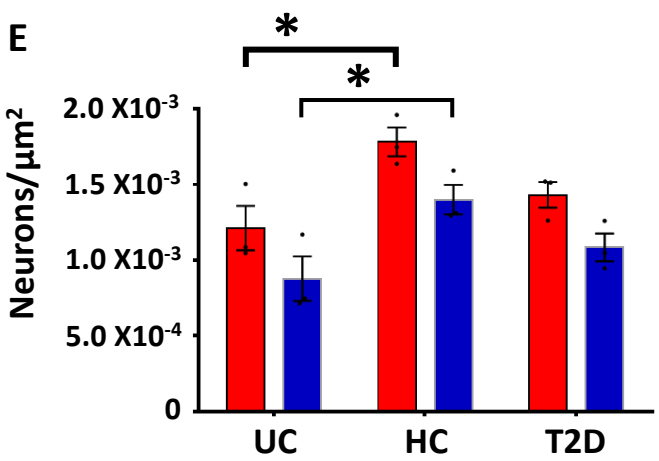


Male T2D Patient's Fecal Filtrates increased Ca^{2+} transient amplitudes in ChAT neurons

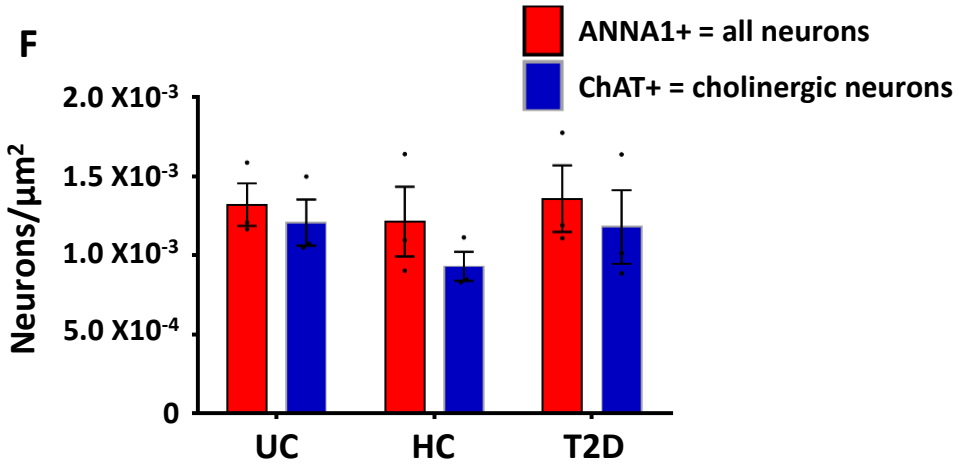
Sex Matters: Sex-Related Differences



Male Recipients



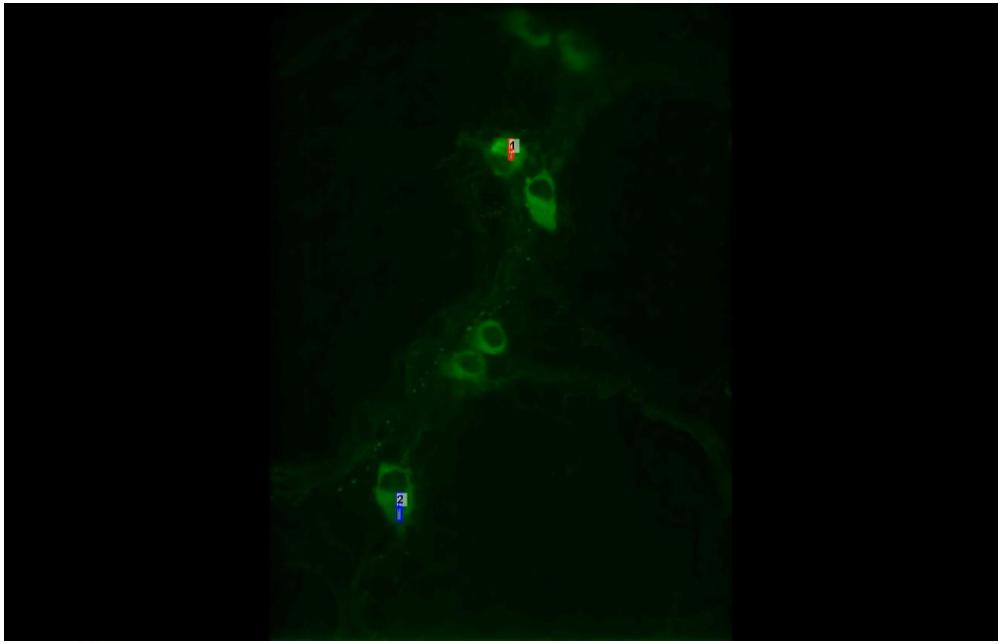
Female recipients

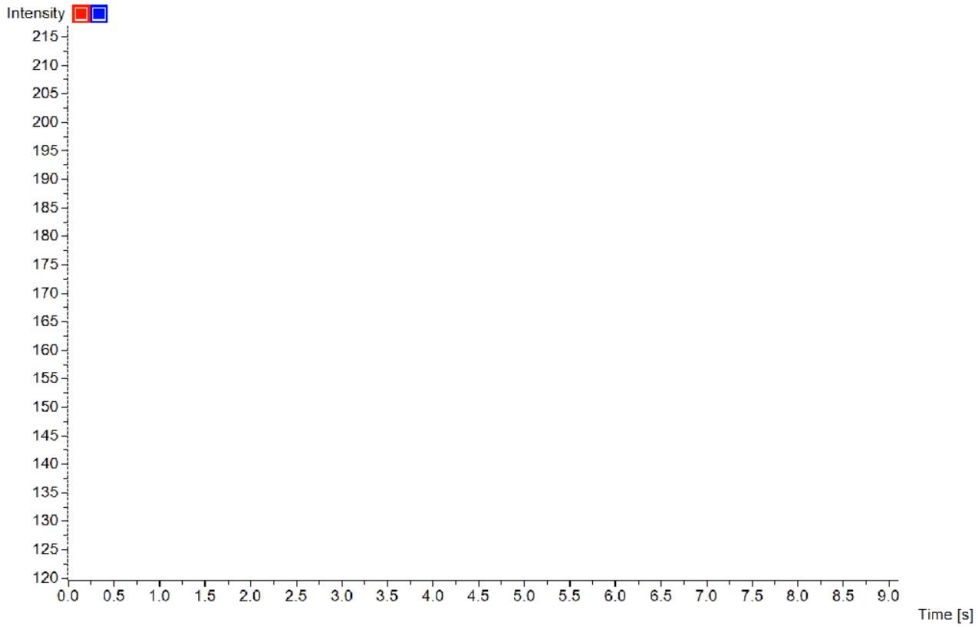


Male T2D Patient's Fecal Filtrates increased ChAT-positive neurons

Voller, et al., Neurogastroenterol Motil. 2026 Jul;38(7):e70401. doi: 10.1111/nmo.70401

Effect on the Activity of nitrergic Myenteric Neurons

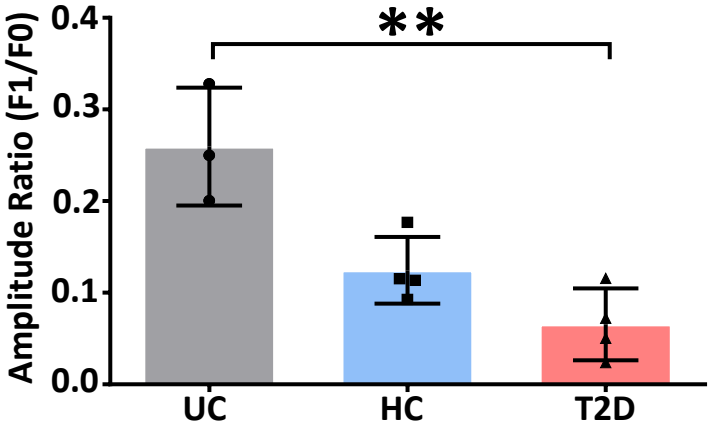


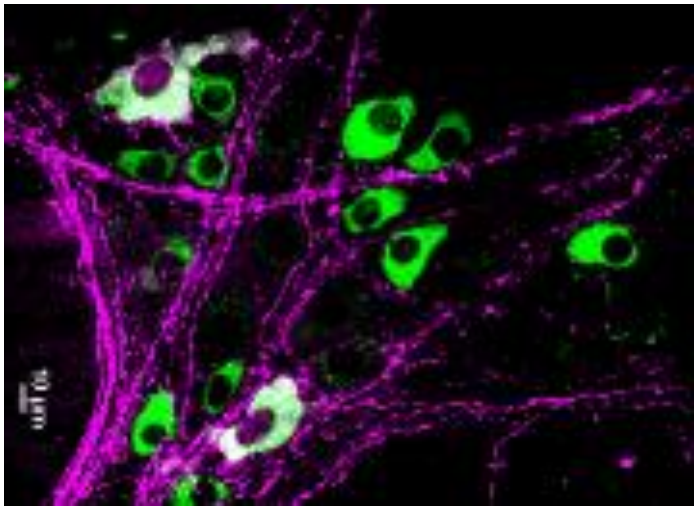


- Ca^{2+} binding protein GCaMP6f in nitrergic myenteric neurons

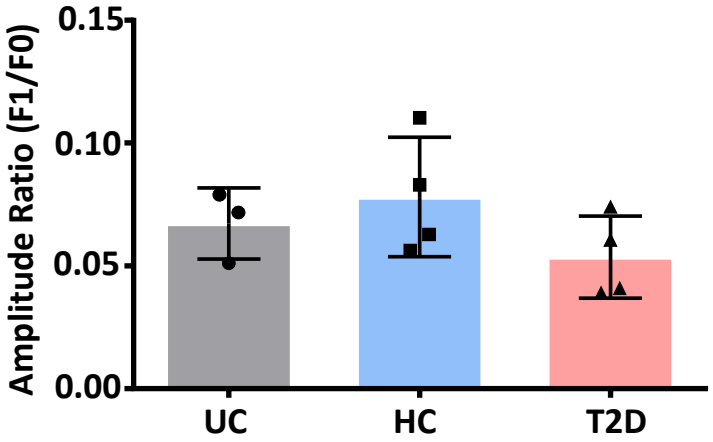
Male T2D Patients Fecal Filtrates Reduced Activity of nitrergic Myenteric Neurons

Stomach, all nitrergic neurons

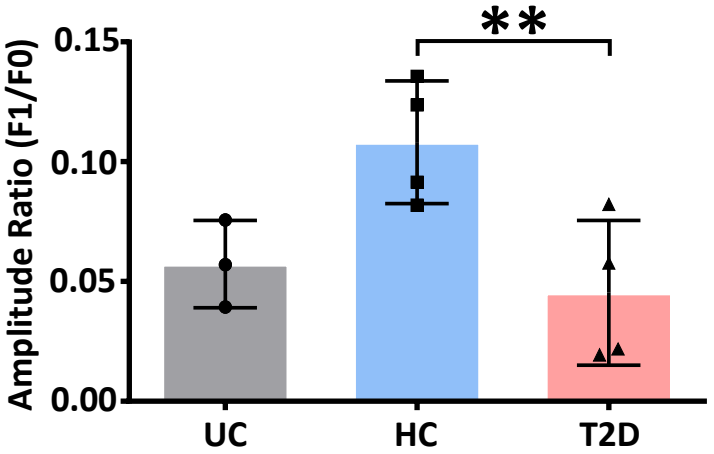




Distal Colon, all nitrergic neurons



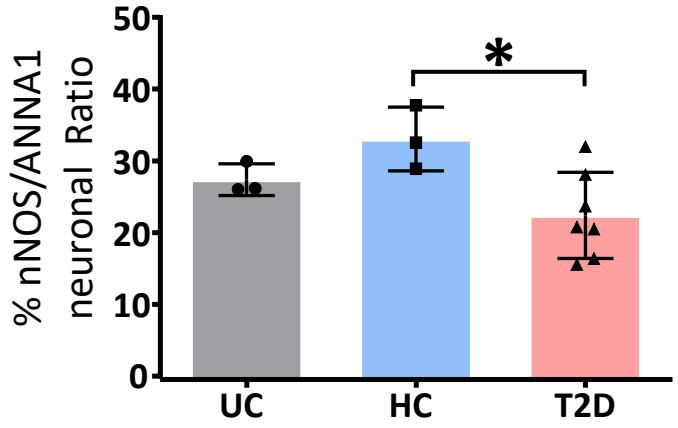
Distal Colon, nitrergic interneurons



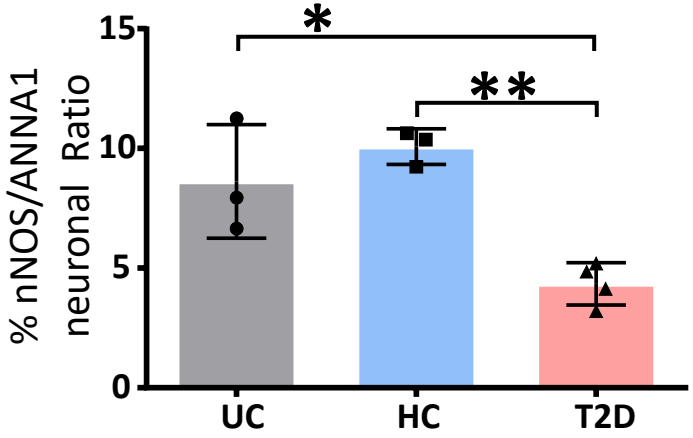
Male T2D patient's fecal filtrates decreased Ca^{2+} activity in nitrergic neurons

Male T2D Patients Fecal Filtrates Reduced the Number of Nitrergic Myenteric Neurons

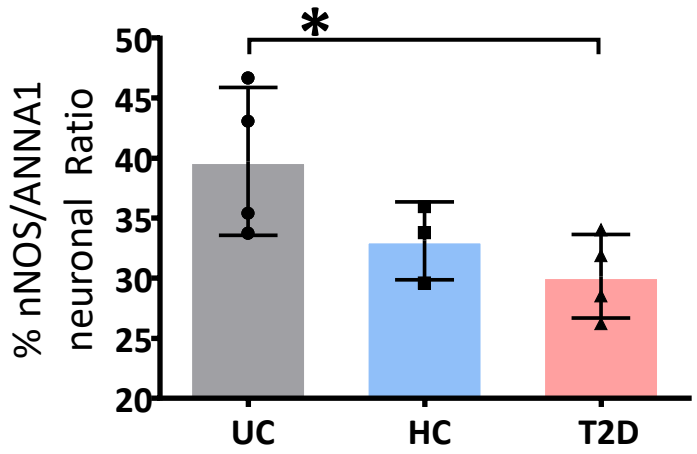
Stomach, all nitrergic neurons



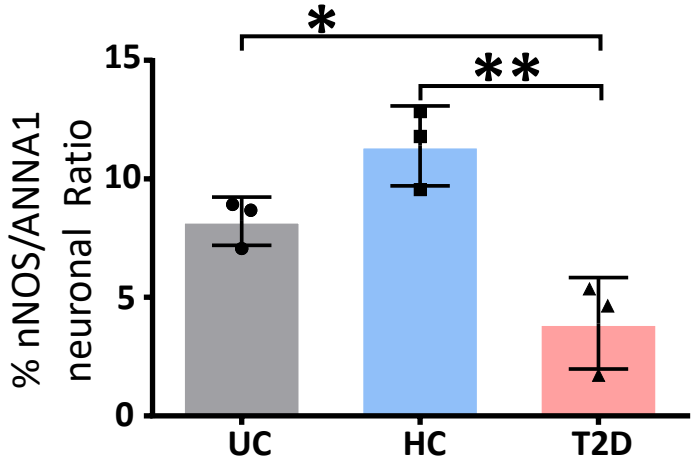
Stomach, nitrergic interneurons



Distal Colon, all nitrergic neurons

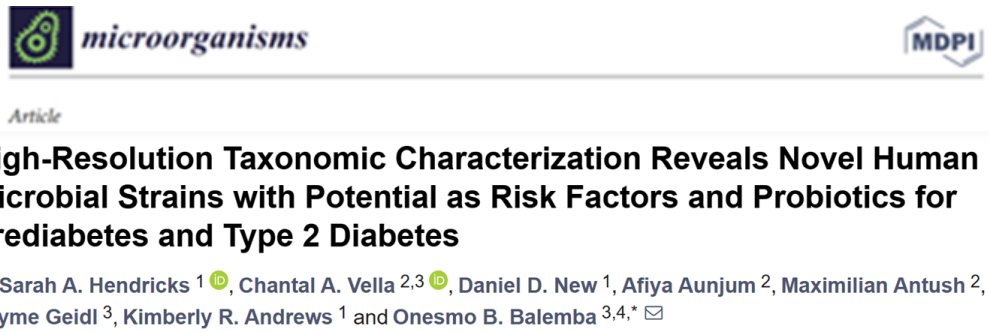


Distal Colon, nitrergic interneurons



Male T2D patient's fecal filtrates decreased nitrergic neurons in the stomach and colon

Which Bacteria Could Be Linked to Harmful Molecules?



microorganisms

MDPI

Article

High-Resolution Taxonomic Characterization Reveals Novel Human Microbial Strains with Potential as Risk Factors and Probiotics for Prediabetes and Type 2 Diabetes

by Sarah A. Hendricks [1], Chantal A. Vella [2,3], Daniel D. New [1], Afiya Aunjum [2], Maximilian Antush [2], Rayme Geidl [3], Kimberly R. Andrews [1] and Onesmo B. Balemba [3,4,*]

- Full-length 16S-ITS-23S sequencing (**INTUS** BIOSCIENCES)

Several bacterial taxa increased in abundance with progression from prediabetes to T2D:

- *Odoribactor laneus,*
- *Flavonifractor plautii,*
- Bacteroides, including *Bacteroides vulgatus (presently Phocaeicola vulgatus)*

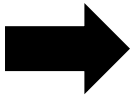
Hendricks S, et al., Microorganisms 2023, 11(3), 758

Which Bacteria Could Be Linked to Harmful Molecules?

- Dr. Mark Driscoll and Dr. Daniel Fasulo (Intus Biosciences LLC)

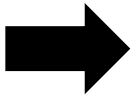
- They performed data mining using our long-amplicon (16S-ITS-23S) sequencing data to identify bacterial strains linked to T2D

Phocaeicola vulgatus whole genomes from NCBI



INTUS BIOSCIENCES

Daydreamer Software



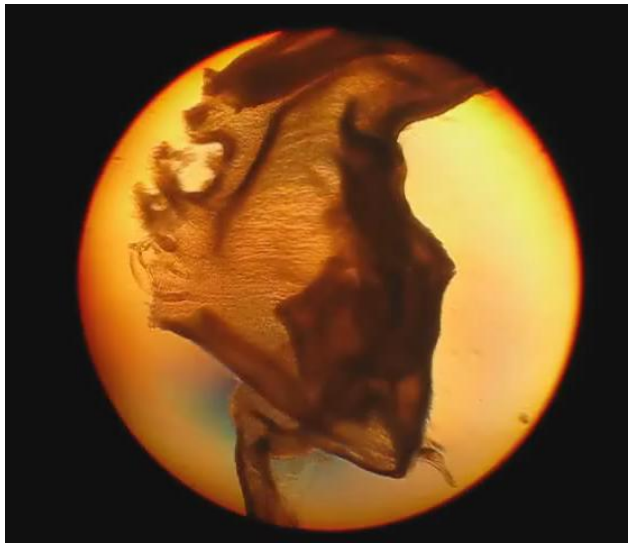
One *P. vulgatus* variant occurs in healthy people but is more abundant in those with T2D

16S Variant: ACCACGCGAGTGGATGCCAATCCC**[C|A]**AAAACCTCTCTCAGTTCGGACTGG

Which Bacteria are Associated with The Injurious Molecules?

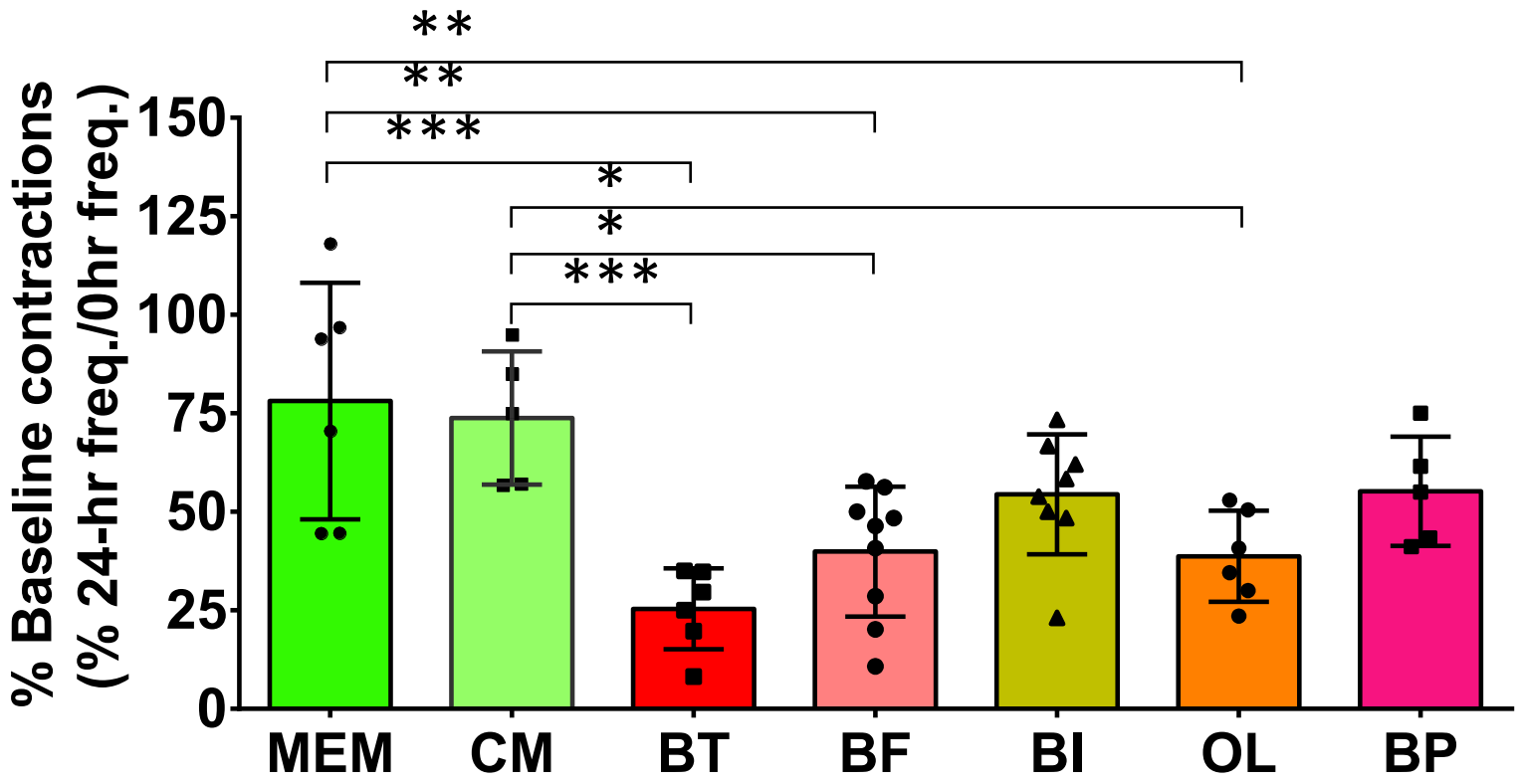
- Tested filter-sterilized (0.2µm) bacterial culture supernatants at 1:10 dilution

Bacterial Strain	**Lab abbreviation**
Odoribactor laneus	OL
Bacteroides thetaiotaomicron	BT
Bacteroides fragilis	BF
Bacteroides intestinalis	BI
Blautia producta	BP
Control: MEM Hepes	MEM
Control: Bacterial culture medium	CM



Bacteria source: The Leibniz Institute DSMZ - German Collection of Microorganisms and Cell Cultures GmbH

Which Bacteria are Associated with The Injurious Molecules?

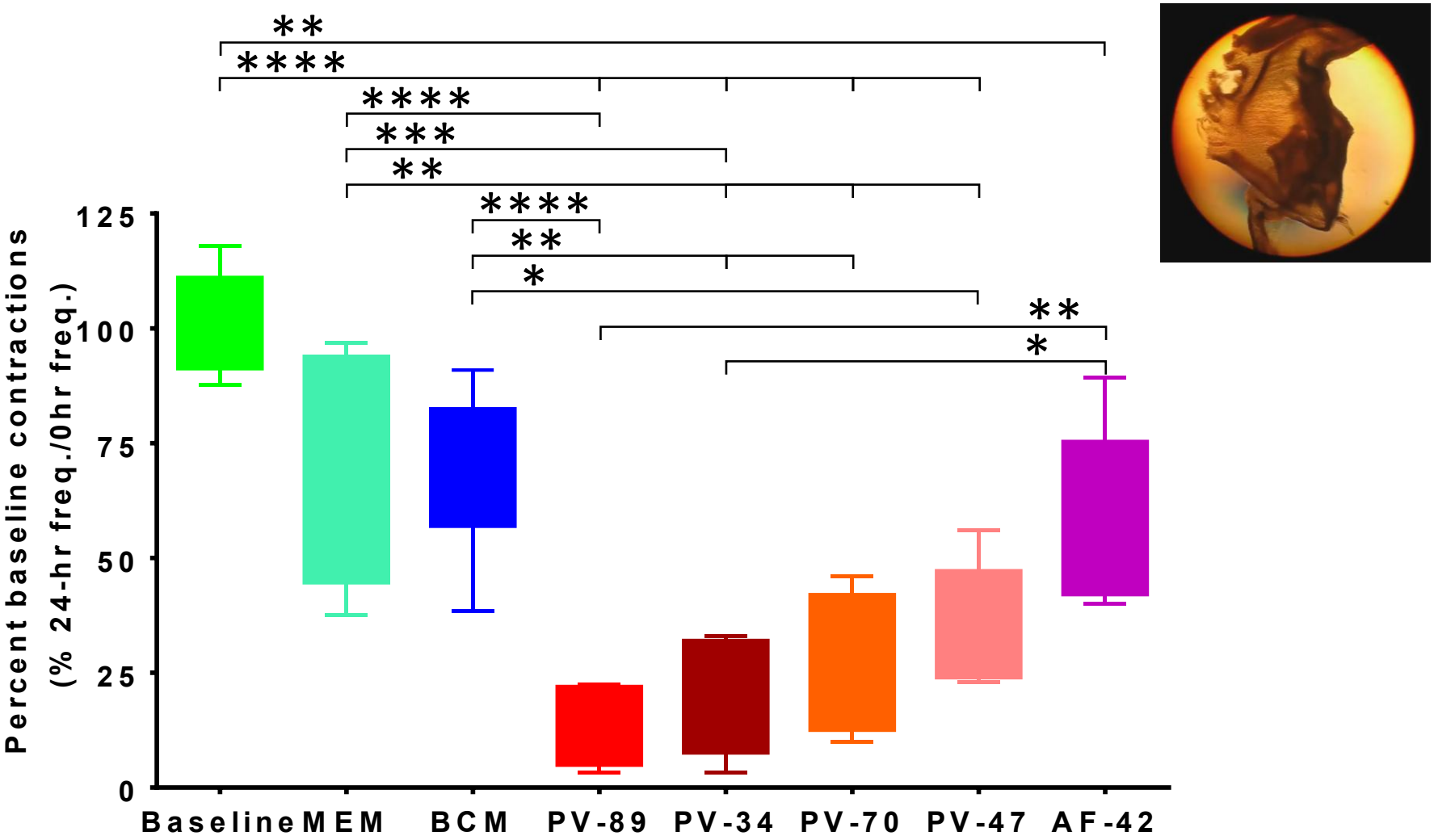


- Results suggest marked strain-level differences in bacterial byproducts that reduce gut contractility

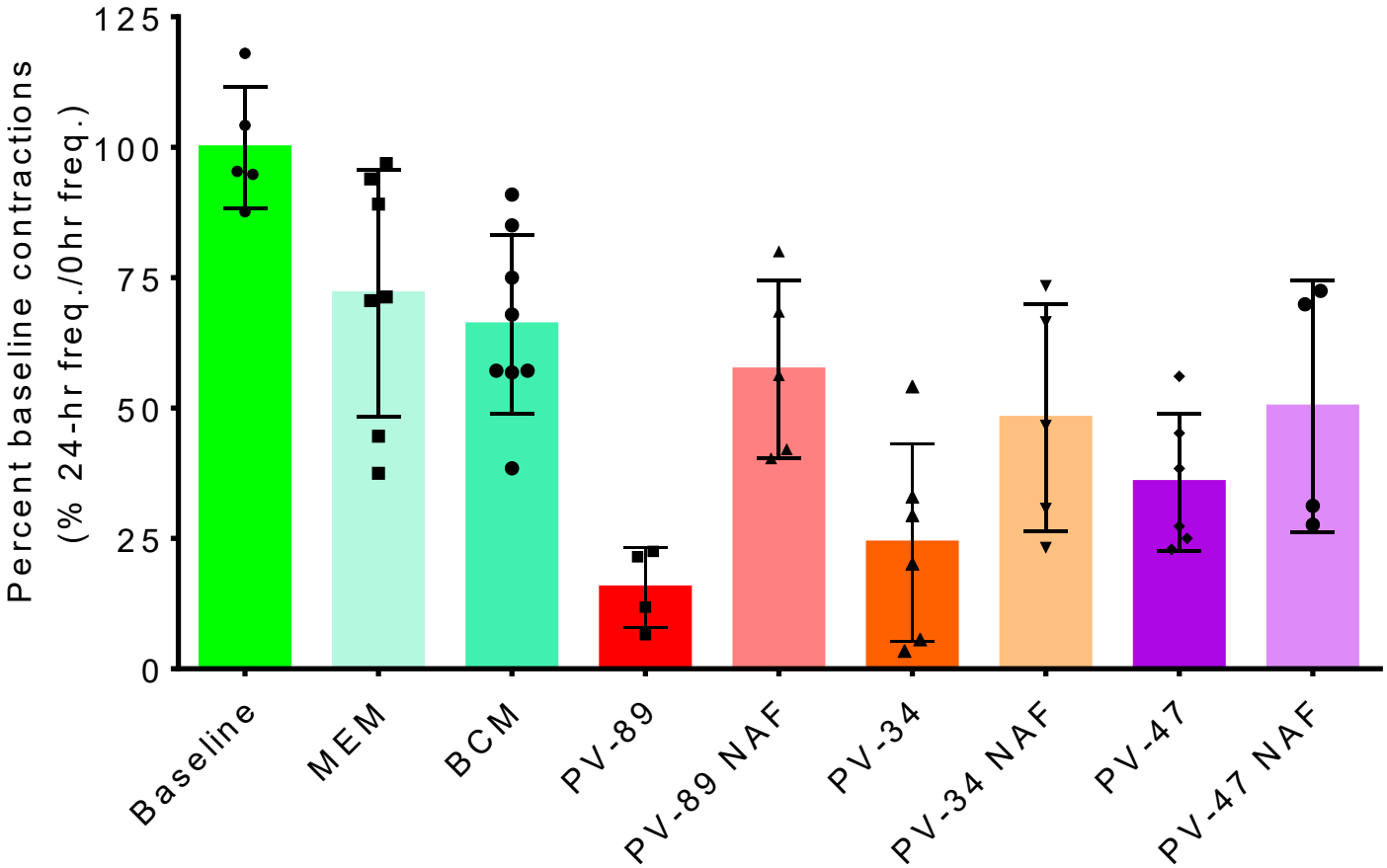
Do *Phocaeicola vulgatus* Byproducts Instigate ENS Neuropathy and Dysmotility?

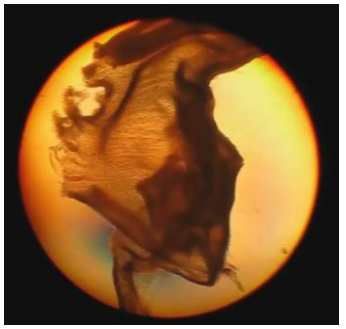
DSMZ ID	***Species***	**Lab Abbreviation**
DSMZ 3289	*Phocaeicola vulgatus*	PV-89
DSMZ 108234	*Phocaeicola vulgatus*	PV-34
DSMZ 1447	*Phocaeicola vulgatus*	PV-47
DSMZ 9670	*Phocaeicola vulgatus*	PV-70
DSMZ 17242	*Alistipes finegoldii*	AF-42
Control	Bacteria culture medium	BCM
Control	Tissue culture medium	MEM

P. Vulgatus DSM-3289 Supernatants Dramatically Decreased Intestinal Muscle Contractions



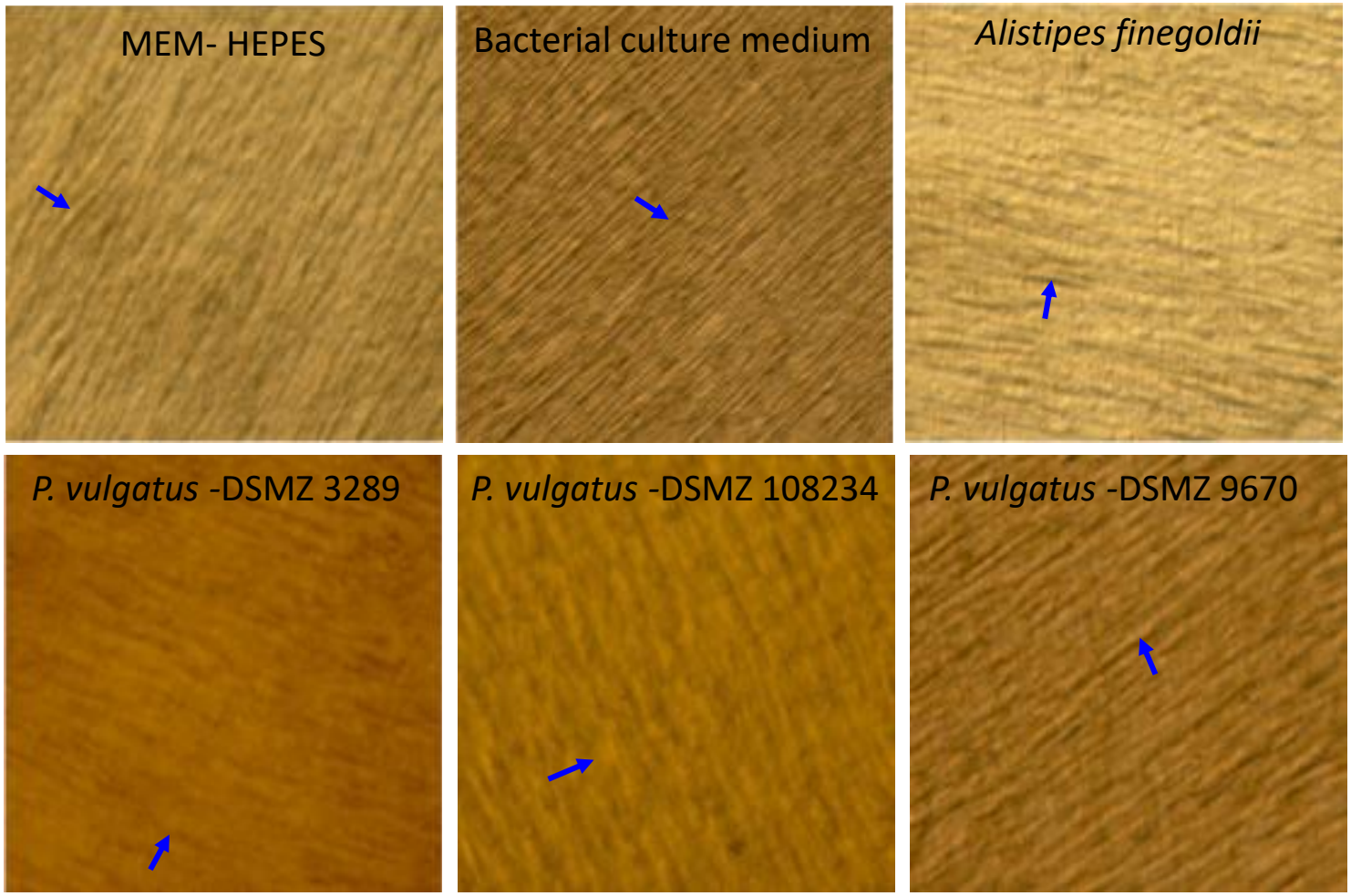
Serine Proteases Mediate Anti-contractility Effects of PV-89, PV-34, and PV-47 supernatants



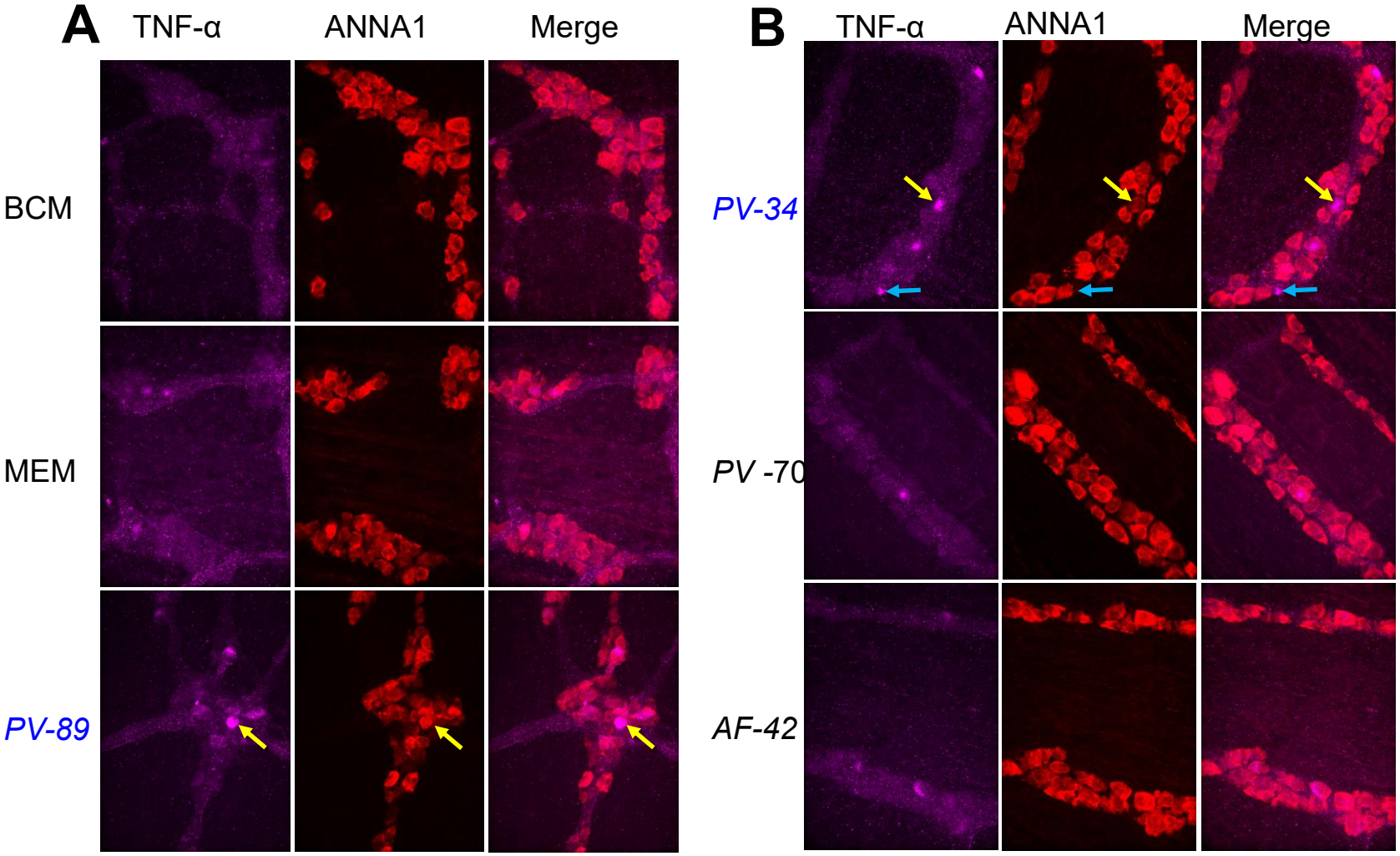


Nafamostat mesylate (NAF; 100 µM) attenuated the anti-contractility potency of PV-89, PV-34, and PV-47 supernatants (n = 3-6 mice).

P. vulgatus Strains PV-89 and PV-34 Caused much Smooth Muscle Swelling than PV-47



Specific *P. vulgatus* Strains Activate TNF-α in the ENS



Culture supernatants of PV-89 and PV-34 activated TNF-α in myenteric neurons (yellow arrows) and enteric glial cells (light blue arrows). ANNA1 is a marker of all neurons.

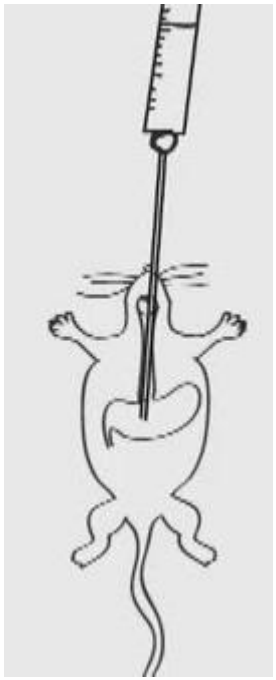
Do Specific *P. vulgatus S*trains Activate Neuroinflammation *in vivo?*

PV-89, PV-34, PV-47, and PV-70 culture supernatants and cell lysates were transplanted into healthy mice

1:10 dil.
X3 @ week
4 weeks

Male mice recipients

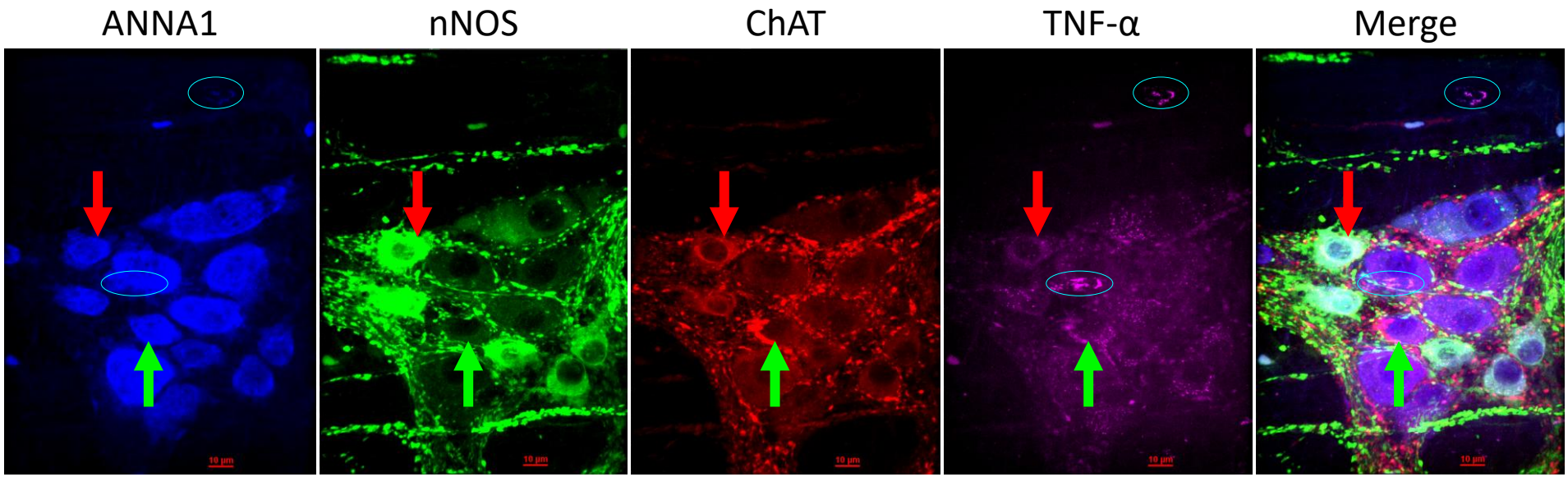




Control: saline or AF-42 supernatants/lysates

2. *Central Image source: JoVE Journal Neuroscience. doi: 10.3791/2123*

P. Vulgatus Strain DSMZ-89, DSMZ-34 Activated Neuroinflammation in Mouse Stomach



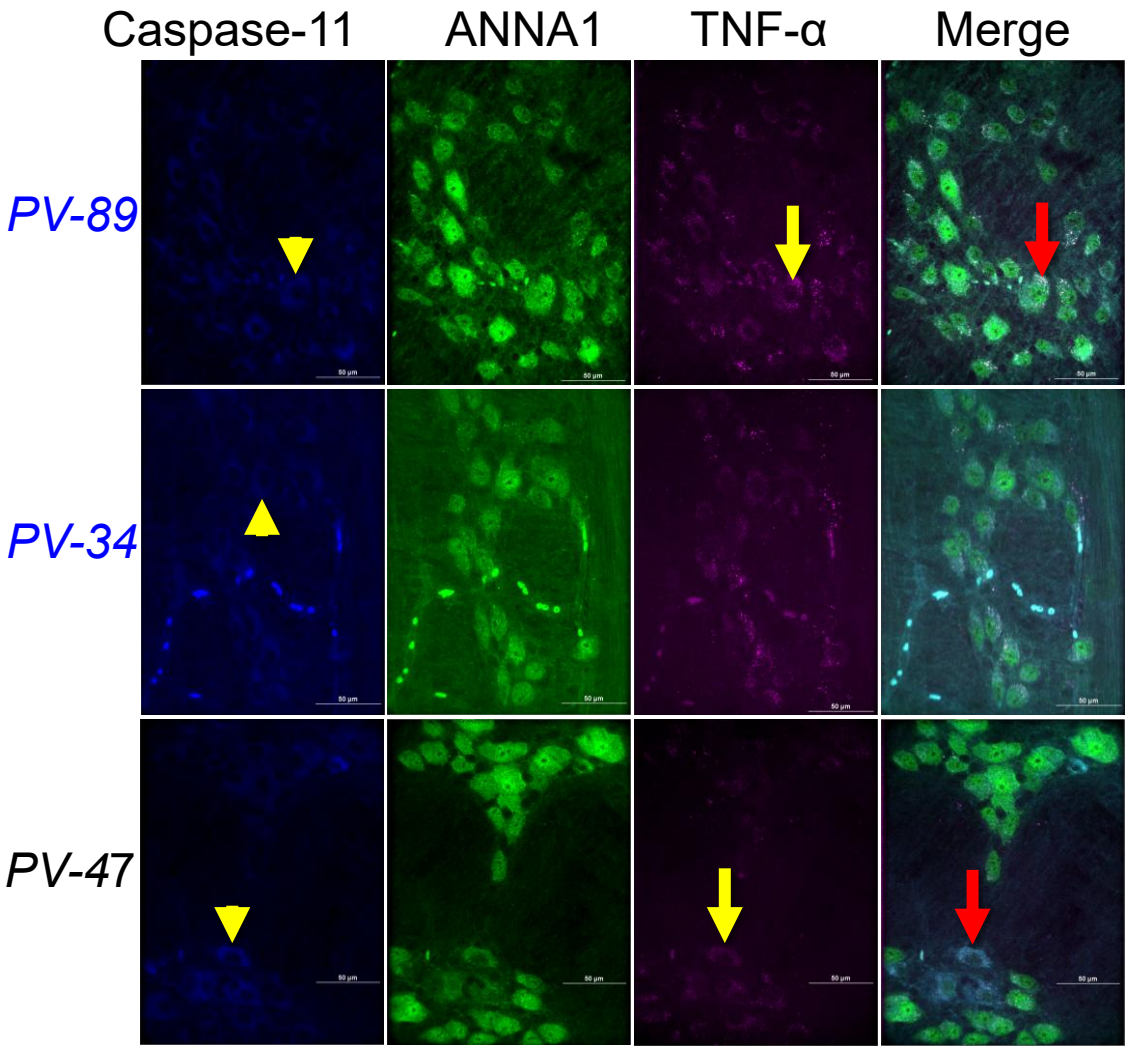
Stomach myenteric ganglion

Red arrows: nNOS, ChAT and TNF-α labeled neuron

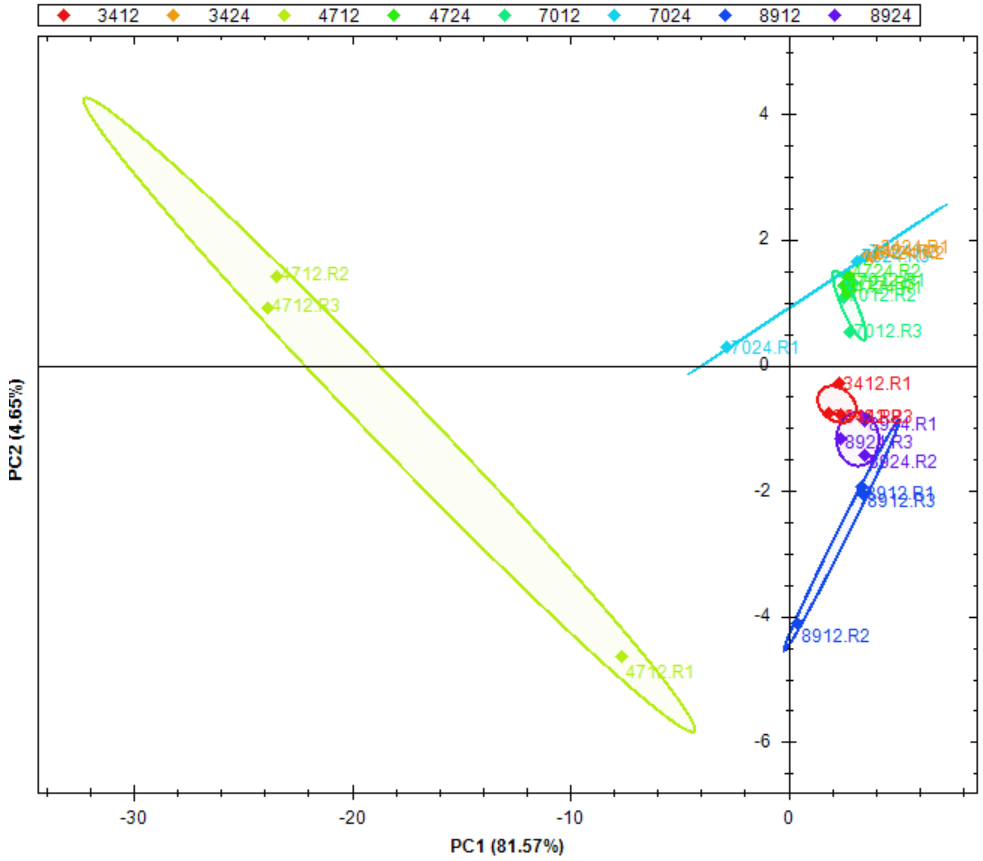
Green arrows: nNOS negative, ChAT and TNF-α positive neuron

Aqua elliptical: immune cells

P. Vulgatus Strain DSMZ-89, DSMZ-34 Activated Caspases + TNF-α in the Stomach's Myenteric neurons



Principal Component Analysis Suggests Variation in Protein Byproducts



PCA plot, *Phocaeicola vulgatus* (PV-89, PV-34, PV-47, and PV-70) culture supernatants

Conclusions

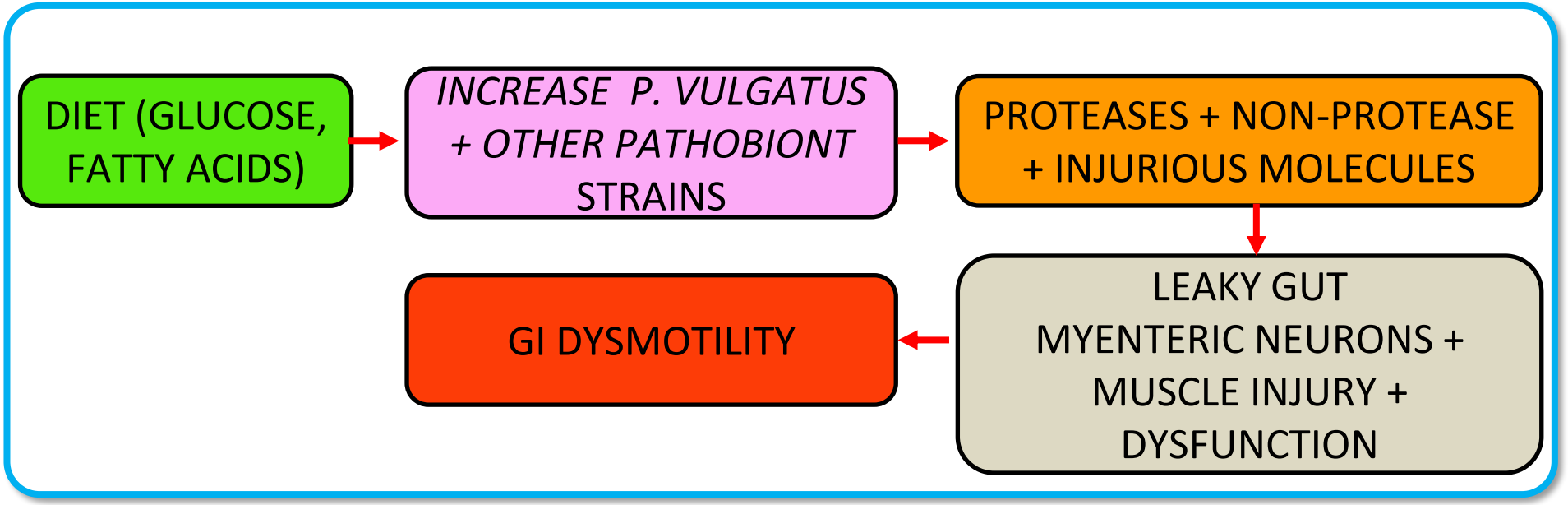
1. Damage to the “gut brain” (ENS neuropathy) occurs before impaired glucose tolerance

2. ENS neuropathy and dysmotility are triggered by “toxic” molecules solubilized in fecal fluid

3. They include proteases, microbial endotoxins, and yet-to-be-identified molecules

4. Byproducts from specific *Phocaeicola vulgatus strains* (DSMZ 3289 and DSMZ 108234) could be contributing to damaging enteric neurons and glial cells during T2D development

Conclusions cont..



Future research

1. Identify supernatant's culprit molecules using proteomics and metabolomics approaches

2. Identify other members of dysbiotic bacterial communities that can trigger ENS neuropathy and dysmotility

3. Determine the mechanisms of action